Pre-Qualification Submission

PSDP Project — Central Monitoring Unit (CMU)

for IPTV Streams of 124 Channels

Submitted to:

Directorate of Electronic Media & Publications (DEMP)

Ministry of Information & Broadcasting

Government of Pakistan, Islamabad

Tender Reference: P28901

Submission Deadline: Monday, 18 May 2026, 11:00 AM (PKT)

Channel: e-PADS v2.0 (EPADS)

Submitted by:

BesTel Communications (Pvt) Ltd.

Pakistan | USA | KSA | UK | Spain

Website: https://bestel.tech

Date: May 17, 2026

BesTel Communications (Pvt) Ltd.

May 16, 2026

To:
The Assistant Director
Directorate of Electronic Media & Publications (DEMP)
B.F Building, Zero Point
Islamabad, Pakistan

Subject: Submission of Pre-Qualification Application for "PSDP Project Central Monitoring Unit (CMU) for IPTV Streams" (Tender Ref: P28901)

Dear Sir,

BesTel is pleased to submit its pre-qualification application for the establishment of the **Central Monitoring Unit (CMU) for IPTV Streams** (Tender Ref: P28901). We have thoroughly reviewed the Pre-Qualification Document (including all instructions, evaluation criteria, and the indicative scope of work) and confirm our compliance with all mandatory requirements.

BesTel's flagship, in-house developed software **Analysys.tech** has been successfully deployed for multiple clients since **2014** – a continuous operational track record spanning over a decade. This platform is currently serving multiple government, defense, and private sector organizations, demonstrating its maturity, security, and reliability.

Specific to this CMU requirement, Analysys offers:

- **Multi-Source Ingest & Failover:** Our solution ingests TV channel feeds from multiple sources (satellite, IP, fiber). As highlighted in your scope (Section 3.b – Multi-Source Automatic Failover), our failover server automatically switches to the best quality stream in real-time, ensuring **high availability of all 124+ channels** under any network or source condition.
- **API-First Architecture:** Analysys.tech is built around secure REST-based APIs, which exactly matches your requirement (Section 3.c) for exposing live streams and metadata to the Ministry's AI and monitoring systems.
- **24/7 Operations:** We are fully prepared to meet the 24/7 monitoring, logging, and SLA compliance requirements.

BesTel Communications (Pvt) Ltd.

To further strengthen our technical and operational capability, BesTel is supported by **KK Networks** – one of the country's largest broadband and IPTV providers with **14 Network Operations Centers (NOCs)** nationwide. Their backbone infrastructure, last-mile expertise, and IPTV headend experience will provide unmatched redundancy, network resilience, and on-ground support for the CMU. All relevant partnership documents and KK Networks' profile are included in our submission.

Based on your PDS (Page 20-22) and Evaluation Criteria (Page 23-24, 36-39), we confirm that BesTel, together with KKN, fully meets or exceeds all the evaluation criteria. We also confirm that all information provided in our submission is true and correct.

We look forward to being pre-qualified and participating in the subsequent bidding stage.

Thank you for your consideration.

Sincerely,

Waseem A. Rana
CEO
BesTel Communications (Pvt) Ltd
0300 841 2244
waseem.rana@bestel.tech

To,
Assistant Director
Directorate of Electronic Media & Publications
(DEMP)
BF Building, Zero Point, Islamabad
Pakistan

Date: May 15, 2026

Subject: Prequalification for PSDP Project Central Monitoring Unit "CMU" for IPTV Streams

Dear Sir,

This is to certify that KK Networks is a nationwide provider of high-speed broadband and allied FTTH services to both corporate and domestic users. KK Networks maintains extensive infrastructure across all 14 telecom regions of Pakistan to support its services to its clients.

KK Networks has been providing high-speed connectivity, hosting, and IPTV services to BesTel Communications (Pvt) Ltd since 2022, and KK Networks undertakes to continue providing these services to BesTel for the subject project.

KK Networks owns and maintains its own as well as leased extensive optical fiber infrastructure, data centers, headends, satellite receivers, and other critical assets to deliver quality services with minimal interruption and downtime. KK Networks has ensured that there is no single point of failure in its services, guaranteeing 99.9% availability.

For and on behalf of
KK Networks (Private) Limited

Authorized Signatory

Name: Imran Ali

Designation: CEO

Company Stamp



1. Executive Summary

BesTel Communications (Pvt) Ltd. specializes in designing, supplying, installing, integrating, commissioning, and supporting a complete **Central Monitoring Unit (CMU) for IPTV Streams** for government and private sectors. Our AI based platform, Analysys, continuously ingests, transcodes and streams **200+ national and international television channels** on a 24×7 basis, and expose those streams plus associated metadata as secure, documented REST APIs for downstream AI-enabled monitoring and analytics consumption.

Analysys is BesTel's production-grade IPTV intelligence platform (analysys.tech), which is already operating at parallel scale (200+ channels) for multiple clients and incorporates several AI capabilities that are uniquely aligned with the long-term objectives of DEMP, including:

- **ContentSense** — automated content classification and topic discovery across live streams.
- **InstaClip** — on-demand clip generation from any monitored channel within seconds.
- **TVC Tracking** — commercial detection and tracking with measured accuracy of 99.9%.
- **Speech-to-Text** — multilingual transcription (English/Urdu) of broadcast audio for search and analytics.

The architecture is designed around the following non-negotiable principles, which are also reflected in the DEMP RFP:

- **Carrier-grade resilience:** multi-source automatic failover with a minimum of three (3) independent acquisition sources per channel.
- **Standards-first delivery:** MPEG-TS over IP, RTP/RTSP, RTMP and HLS for ingestion and onward delivery; ABR transcoding for multi-bitrate distribution.
- **Secure API-first exposure:** OAuth 2.0/JWT-secured REST endpoints, fine-grained authorization, rate-limiting and full audit trails.
- **Browser-native operations:** multi-layout video wall with a minimum of six (6) configurable layouts, accessible through any modern browser without proprietary client software.
- **SaaS-capable deployment:** the platform can be delivered either as an on-premise installation at the CMU data center or as a fully managed SaaS, requiring zero hardware from DEMP.
- **24×7 NOC support:** manned Network Operations Centre with documented escalation matrix, incident response, and continuous performance monitoring.

In order to enhance its capabilities to deliver this project in the best possible fashion, **BesTel has collaborated with KK Networks (Pvt) Limited**, a well-known and highly technical broadband and IPTV services provider in Pakistan. **Besides KK Networks, BesTel has acquired streams from other IPTV services providers. BesTel also possibly acquire streams from PTCL and NayaTel.**

BesTel commits to a **fifteen (15) calendar day** core deployment cycle from contract effective date to operational go-live, followed by a **fifteen (15) day** stabilization and acceptance period, and ongoing operations, maintenance, and 24×7 support for the full contract tenure. The phased schedule, milestones and acceptance gates are detailed in Section 10 of this proposal.

2. About BesTel

A. Company Overview

BesTel Communications (Pvt) Ltd. is a Pakistan-incorporated technology company providing carrier-grade software platforms and managed services to government, telecom, broadcasting, and enterprise customers. With operational presence across **Pakistan, the United Kingdom, the Kingdom of Saudi Arabia, the United States and Spain**, BesTel combines local execution capacity with international engineering depth.

The company specializes in:

- Media monitoring and broadcast intelligence platforms (IPTV ingestion, transcoding, archival, search, AI analytics).
- ISP operational software — CRM, billing, provisioning, and TR-069/TR-369 Auto Configuration Servers (ACS).
- Custom enterprise software — full-stack web platforms, microservices, real-time streaming back-ends.
- Network infrastructure design, deployment and security hardening for service providers and government clients.
- Embedded systems and IoT for telecommunications and broadcasting use-cases.

B. Flagship Platform — Analysys

Analysys (web.analysys.tech) is BesTel's production IPTV intelligence platform, purpose-built for the kind of centralized broadcast monitoring DEMP is now seeking to formalize under the PSDP project. The platform is already operational with a Government of Pakistan client across 125 channels and combines four functional layers:

- **Acquisition:** multi-source ingestion (satellite/IRD, IP/multicast, OTT/HLS) with automatic failover.
- **Streaming:** FFmpeg-based real-time transcoding (libx264 ultrafast + AAC), MediaMTX RTMP/HLS distribution, hls.js browser playback, multi-quality ABR rendition.
- **Storage & search:** up to 250 TB rolling archive on QNAP NAS clusters, MongoDB metadata indexing, timestamped clip retrieval.
- **AI analytics:** ContentSense, InstaClip, TVC Tracking (99.9%) and speech-to-text for Urdu/English broadcasts.

The architecture, codebase and operations playbook proposed for the CMU project are direct evolutions of this proven Analysys stack, which materially reduces delivery risk and shortens time-to-go-live.

C. Why BesTel — Differentiators for DEMP

- **Existing GoP track record:** BesTel is already delivering an analogous, larger-scale (125 channel) IPTV intelligence service to multiple government, defense, and private clients, including a deployment success record on file with DEMP.
- **Proven 99.9% accuracy:** our TVC tracking and content classification engines have measured production accuracy of 99.9%, providing DEMP with a credible AI layer for downstream applications.
- **Standards-first:** HLS, MPEG-TS over IP, RTP/RTSP, REST/JSON, OAuth 2.0 — zero proprietary lock-in.
- **Smart Deployment:** BesTel offers SaaS-hosted access to its flagship platform Analysys. This means quickest deployment, no hardware at DEMP side, no software issues, no responsibility on DEMP to maintain the system and access to smart features of Analysys. DEMP can access Analysys platform from any system connected on the internet.
- **24×7 NOC:** Pakistan-based engineers, no time-zone or escalation friction.
- **End-to-end ownership:** BesTel performs design, supply, integration, training and support in-house — not as a re-seller — ensuring single-point accountability.

D. Core Competencies

- **Artificial Intelligence and Machine Learning**: Expertise in developing AI-powered solutions for data analysis, pattern recognition, and predictive modelling.
- **Digital Forensics and Cybersecurity**: Specialized in forensic tools, secure system design, and incident response solutions.
- **Software Development**: Custom application development with expertise in mobile, web, and desktop technologies.
- **OSINT and Intelligence Tools**: Development of specialized intelligence gathering and analysis platforms for law enforcement and security agencies.
- **Government and Enterprise Solutions**: Secure solutions for government agencies with experience in handling sensitive data and compliance requirements.

3. Understanding of Requirements

BesTel has carefully studied the Pre-Qualification Document issued by DEMP and the embedded Request for Proposals (RFP) for the PSDP Project Central Monitoring Unit (CMU) for IPTV Streams of 124 Channels. Our reading of the requirement is summarized below to demonstrate alignment between DEMP's expectations and our proposed solution.

A. Project Objective

DEMP intends to establish a centralized IPTV streaming and monitoring facility (the CMU) that continuously ingests licensed feeds of 124 national and international television channels and exposes them as secure, documented APIs for onward consumption by the Ministry's and attached organizations' AI-enabled monitoring and analytics applications. The platform must operate on a 24×7 basis, with redundancy, disaster recovery, and clear service level compliance.

B. Key Functional Requirements

#	Requirement Area	Our Understanding
a	IPTV acquisition & headend	Provide 124+ live TV feeds to DEMP that have been ingested via satellite, IP or fiber using standard broadcast protocols (RTP, RTSP, HLS, MPEG-TS over IP).
b	Multi-source failover	Minimum three (3) independent sources per channel with automatic, sub-second failover and full audit logging.
c	Real-time encoding	Live transcoding to multiple bitrates/formats with adaptive bitrate (ABR) delivery.
d	Centralized streaming	Clustered streaming servers with load balancing, failover and redundancy.
e	SaaS option	Capability to deliver the entire service as a managed SaaS, requiring no hardware from DEMP.
f	Video wall	Browser-native multi-layout video wall with at least six (6) configurable layouts.
g	Secure REST APIs	Authenticated, authorized, rate-limited REST endpoints exposing streams + metadata (channel, language, category, timestamps).
h	Storage & logging	Short-term buffering/recording per requirement; centralized access, error, uptime, and performance logs.
i	Network & security	Switching/routing/firewall/VLAN design and hardening to protect content and APIs.
j	Operations & support	24×7 first-line support, preventive and corrective maintenance, incident response.
k	Training & documentation	Technical documentation, API documentation, design documents, configuration baselines, and end-user/administrator training.

C. Non-Functional Requirements (inferred)

- **Availability:** Platform-level uptime target of 99.9% (industry baseline for broadcast monitoring); per-channel availability subject to source feed availability.
- **Performance:** End-to-end glass-to-glass latency target of less than 6 seconds on HLS; sub-second on RTMP/low-latency HLS where required.
- **Scalability:** Horizontal scale-out for both ingest and distribution; capacity headroom for additional channels without architectural re-design.
- **Security:** Encryption in transit (TLS 1.2+) and at rest; role-based access control; complete audit trail.
- **Compliance:** Adherence to PPRA Rules 2004 procurement obligations; PEMRA-aligned content handling practices for licensed feeds.
- **Interoperability:** Open standards (HLS, MPEG-TS, REST/JSON, OAuth 2.0/JWT) to support unrestricted future integration.

C

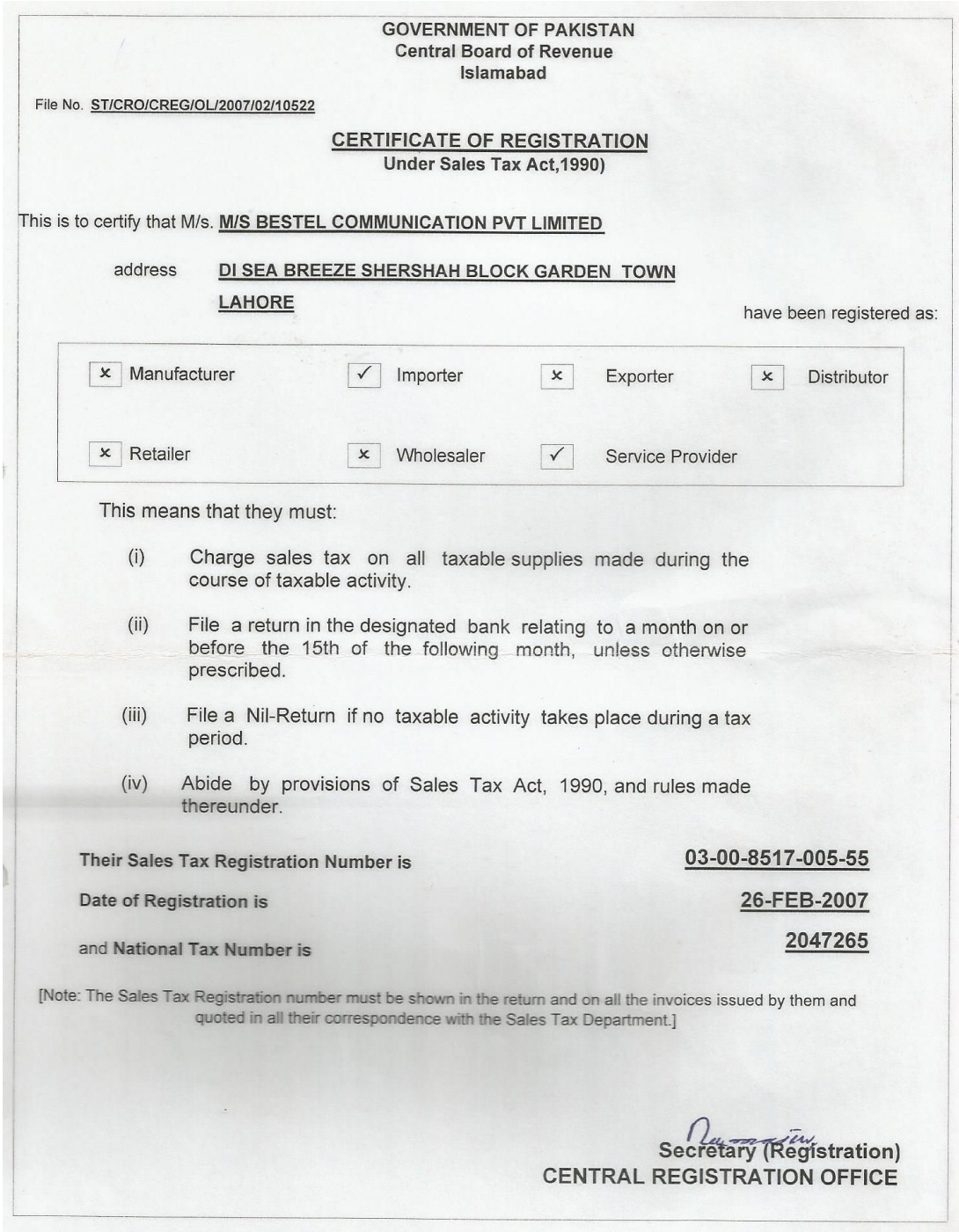
Legal / Regulatory Compliance

A. BesTel Certificates

BesTel Basic Information

Incorporation Certificate	13539/20040502
FBR NTN	2047265
FBR GSTN	2047265
PRA NTN	2047265
PSEB Registration	Z-25-01523
PTCL Registration	4101378

FBR NTN Registration



GOVERNMENT OF PAKISTAN
Central Board of Revenue
Islamabad

File No. ST/CRO/CREG/OL/2007/02/10522

CERTIFICATE OF REGISTRATION
Under Sales Tax Act,1990)

This is to certify that M/s. **M/S BESTEL COMMUNICATION PVT LIMITED**

address **DI SEA BREEZE SHERSHAH BLOCK GARDEN TOWN LAHORE**

have been registered as:

[x] Manufacturer	[✓] Importer	[x] Exporter	[x] Distributor
[x] Retailer	[x] Wholesaler	[✓] Service Provider	

This means that they must:

(i) Charge sales tax on all taxable supplies made during the course of taxable activity.

(ii) File a return in the designated bank relating to a month on or before the 15th of the following month, unless otherwise prescribed.

(iii) File a Nil-Return if no taxable activity takes place during a tax period.

(iv) Abide by provisions of Sales Tax Act, 1990, and rules made thereunder.

Their Sales Tax Registration Number is 03-00-8517-005-55

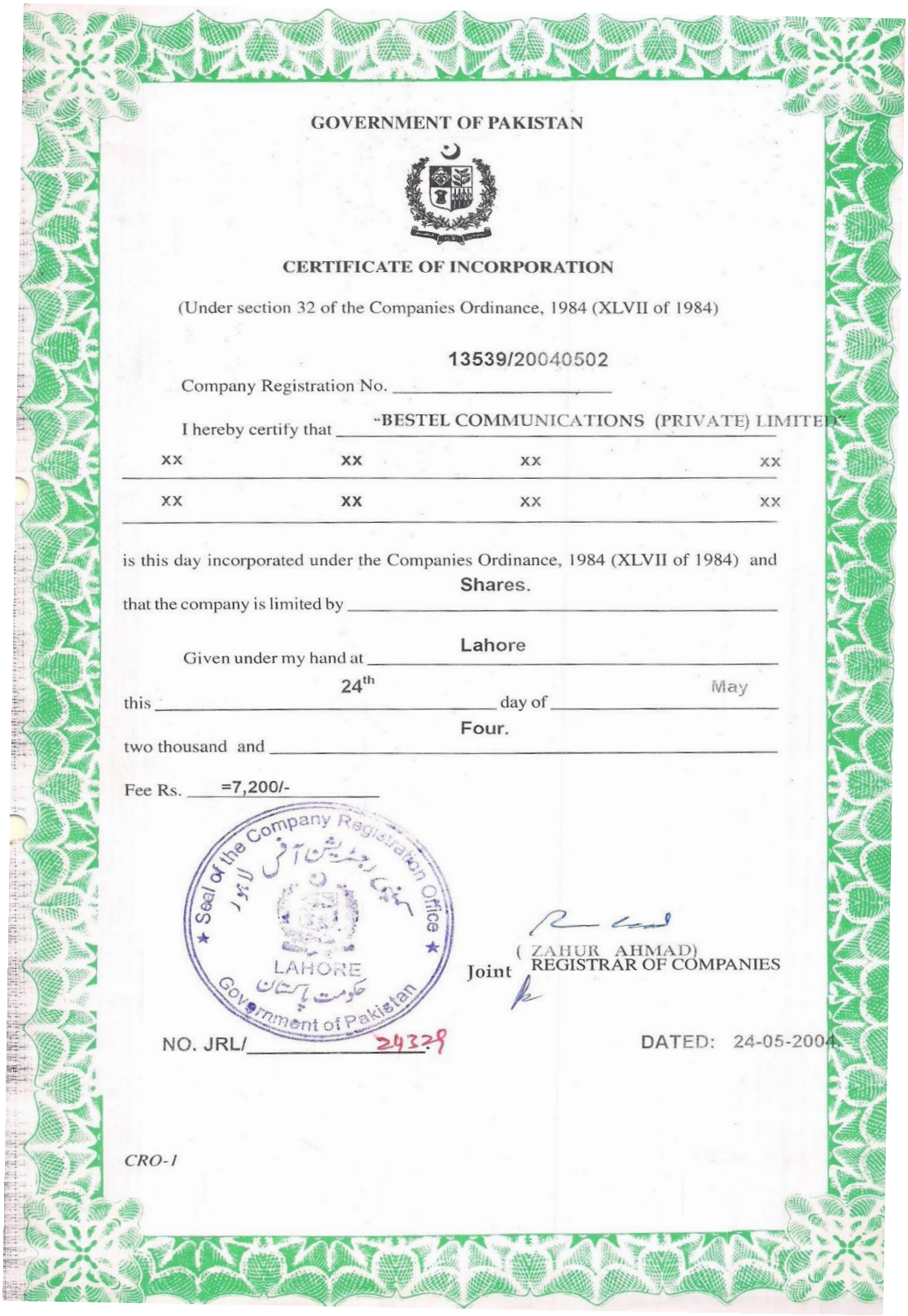
Date of Registration is 26-FEB-2007

and **National Tax Number is** 2047265

[Note: The Sales Tax Registration number must be shown in the return and on all the invoices issued by them and quoted in all their correspondence with the Sales Tax Department.]

Secretary (Registration)
CENTRAL REGISTRATION OFFICE

BesTel Certificate of Incorporation



GOVERNMENT OF PAKISTAN

CERTIFICATE OF INCORPORATION

(Under section 32 of the Companies Ordinance, 1984 (XLVII of 1984)

Company Registration No. 13539/20040502

I hereby certify that "BESTEL COMMUNICATIONS (PRIVATE) LIMITED"

xx xx xx xx

xx xx xx xx

is this day incorporated under the Companies Ordinance, 1984 (XLVII of 1984) and that the company is limited by Shares.

Given under my hand at Lahore

this 24th day of May two thousand and Four.

Fee Rs. =7,200/-

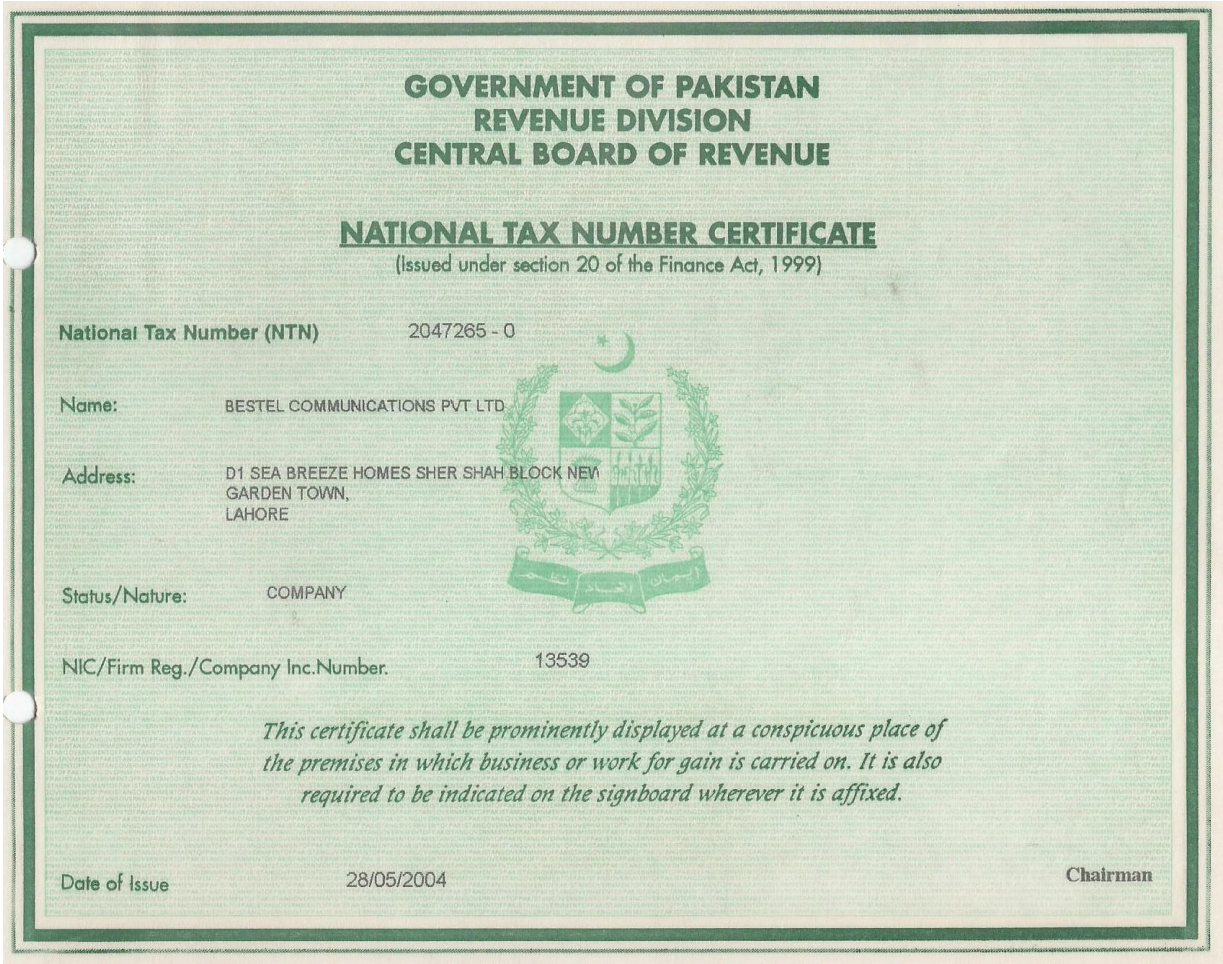
Seal of the Company Registration Office LAHORE Government of Pakistan

Joint REGISTRAR OF COMPANIES (ZAHUR AHMAD)

NO. JRL/ 24329 DATED: 24-05-2004

CRO-1

BesTel National Tax Number



GOVERNMENT OF PAKISTAN
REVENUE DIVISION
CENTRAL BOARD OF REVENUE

NATIONAL TAX NUMBER CERTIFICATE
(Issued under section 20 of the Finance Act, 1999)

National Tax Number (NTN) 2047265 - 0

Name: BESTEL COMMUNICATIONS PVT LTD

Address: D1 SEA BREEZE HOMES SHER SHAH BLOCK NEW GARDEN TOWN, LAHORE

Status/Nature: COMPANY

NIC/Firm Reg./Company Inc.Number. 13539

This certificate shall be prominently displayed at a conspicuous place of the premises in which business or work for gain is carried on. It is also required to be indicated on the signboard wherever it is affixed.

Date of Issue 28/05/2004

Chairman

Federal Board of Revenue
Revenue Division - Government of Pakistan

Online Verification System

Active Taxpayer List (Sales Tax)

Printed On: 13,May, 2026 - 02:57:50 PM

Taxpayer Name:	BESTEL COMMUNICATIONS (PVT.) LIMITED
Registration No.:	2047265
Business Name:	M/S BESTEL COMMUNICATION PVT LIMITED
Current ATL Status:	Taxpayer is Active
Return Filing Status (past 6 months):	The taxpayer e-filed 6 ST&FE Returns during past 6 tax periods

PRA PUNJAB REVENUE AUTHORITY
GATEWAY TO DEVELOPMENT

PRA REGISTRATION CERTIFICATE

2047265

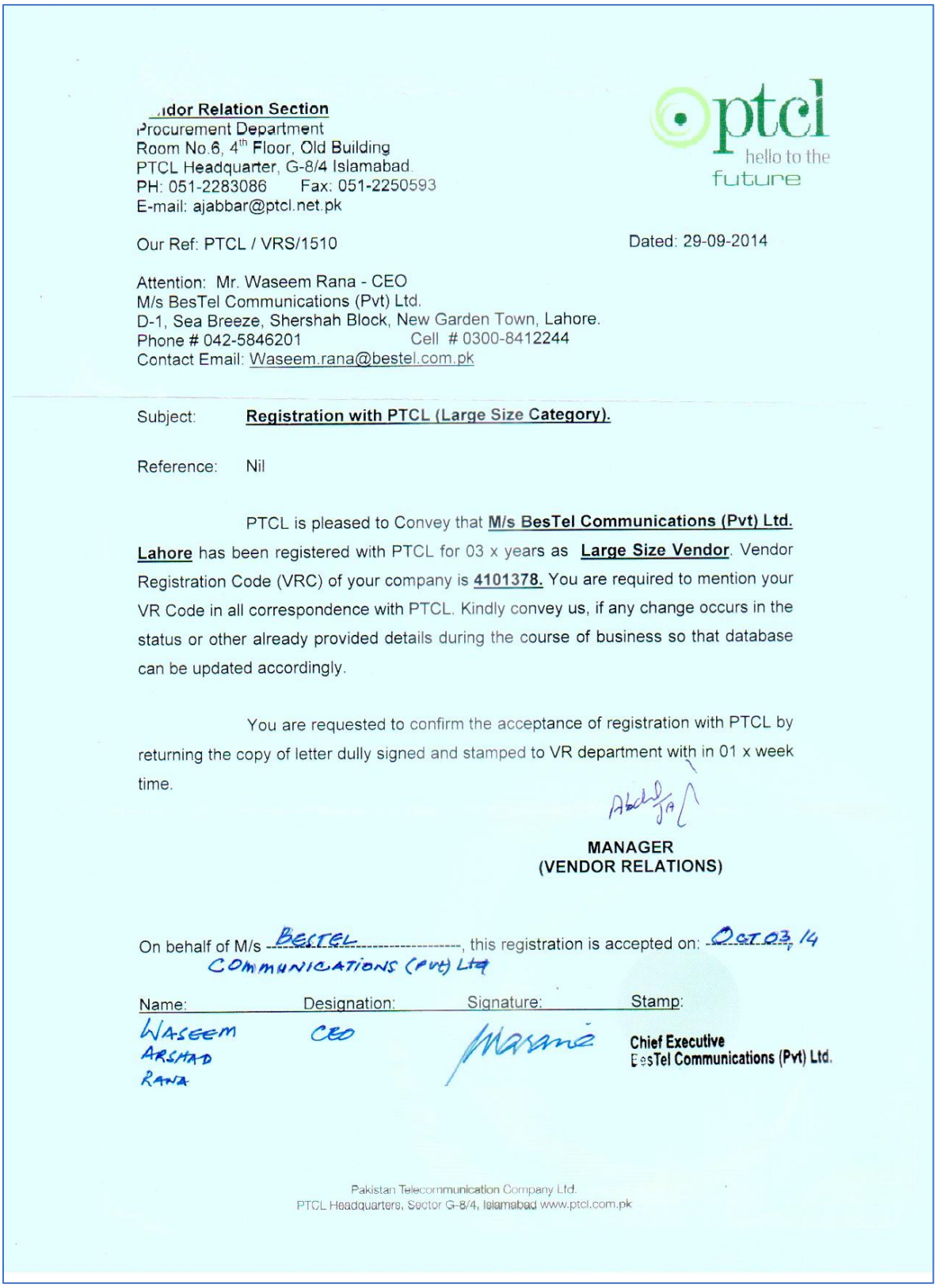
PNTN	2047265-0
Registration No.	2047265
Date of Registration	09-Mar-2018
Name	BESTEL COMMUNICATIONS (PVT.) LIMITED
Type	Company
Business Name	
Business Address	D1 SEA BREEZE HOMES SHER SHAH BLOCK NEW GARDEN TOWN , Lahore , Lahore , Lahore
Category	WithHolding Agent / Service Provider
Inc. /Reg No	00000013539/20040502
Commissionerate	Lahore Commissionerate
Principal Activity	
Other Activities	
Status	Active **As on**

This is a computer generate certificate and, hence, no signatures are required.
This certificate shall be prominently displayed at a conspicuous place of the premises in which business or work for gain is carried on.
NTN certificate is also required to be indicated on the signboard.

042-99205476-77 042-99205716 042-99205481
eSupport@pra.punjab.gov.pk https://e.pra.punjab.gov.pk

Date of Print 17-May-2026

PTCL Registration Certificate



Vendor Relation Section
Procurement Department
Room No.6, 4th Floor, Old Building
PTCL Headquarter, G-8/4 Islamabad.
PH: 051-2283086 Fax: 051-2250593
E-mail: ajabbar@ptcl.net.pk

ptcl hello to the future

Our Ref: PTCL / VRS/1510

Dated: 29-09-2014

Attention: Mr. Waseem Rana - CEO
M/s BesTel Communications (Pvt) Ltd.
D-1, Sea Breeze, Shershah Block, New Garden Town, Lahore.
Phone # 042-5846201 Cell # 0300-8412244
Contact Email: Waseem.rana@bestel.com.pk

Subject: **Registration with PTCL (Large Size Category).**

Reference: Nil

PTCL is pleased to Convey that **M/s BesTel Communications (Pvt) Ltd. Lahore** has been registered with PTCL for 03 x years as **Large Size Vendor**. Vendor Registration Code (VRC) of your company is **4101378.** You are required to mention your VR Code in all correspondence with PTCL. Kindly convey us, if any change occurs in the status or other already provided details during the course of business so that database can be updated accordingly.

You are requested to confirm the acceptance of registration with PTCL by returning the copy of letter dully signed and stamped to VR department with in 01 x week time.

MANAGER
(VENDOR RELATIONS)

On behalf of M/s BESTEL COMMUNICATIONS (PVT) LTD, this registration is accepted on: OCT 03, 14

Name:	Designation:	Signature:	Stamp:
WASEEM ARSHAD RANA	CEO		Chief Executive BesTel Communications (Pvt) Ltd.

Pakistan Telecommunication Company Ltd.
PTCL Headquarters, Sector G-8/4, Islamabad www.ptcl.com.pk

Affidavit / Undertaking Regarding Non-Blacklisting

E-STAMP

ID:	PB-LHR-4A115S5 H9C7E6N83
Type:	Low Denomination
Amount:	Rs 100/-
Description:	CERTIFICATE OR OTHER DOCUMENT-19
Applicant:	Bestel Communication Pvt Ltd [10000-0000000-0]
Representative From:	Bestel Communication Pvt Ltd
Agent:	Mohammad Mujtaba [33203-0338784-7]
Address:	Lahore
Issue Date:	15-MAY-2026 4:09:07 PM
Delisted On/Validity:	22-MAY-2026
Amount in Words:	One Hundred Rupees Only
Reason:	CERTIFICATE OR OTHER DOCUMENT-19
Vendor Information:	Rafiq Ahmad I PB-LHR-1434I Lahore

Scan for online verification

SYED KHALID HASHMI Lic # 495 OATH COMMISSIONER Advocate High Court Lahore

100/ 15/05/2026

ATTESTED SYED KHALID HASHMI OATH COMMISSIONER Advocate High Court Lahore

نوٹ : یہ فرانزیکشن تاریخ اجرا سے سات دن تک کے لیے قابل استعمال ہے ، اس اسٹامپ کی تصدیق بذریعہ ویب سائٹ ، ٹول فری نمبر سے کی جاسکتی ہے۔

CERTIFICATE OF COMPLIANCE

Date: May 14, 2026

To Whom It May Concern,

This is to certify that **BesTel Communications (Pvt) Ltd** is in good legal and professional standing. The company has **never been blacklisted, debarred, or suspended** by any governmental, semi-governmental, or private entity, and has **never been involved in litigation or legal disputes** with any such organization.

Waseem Arshad Rana
CEO
BesTel Communications (Pvt) Ltd

BesTel Communications (Pvt) Ltd

B. KK Network Certificates

KK Network Basic Information

SECP Registration	0094119	25 Jun 2015
FBR NTN	7127176	11 Nov 2015
FBR GSTN	7127176	11 Nov 2015
PRA NTN	7127176-4	9 Jul 2017
PTA Registration	Multiple Licenses	



A009533

SECURITIES AND EXCHANGE COMMISSION OF PAKISTAN

COMPANY REGISTRATION OFFICE, LAHORE

CERTIFICATE OF INCORPORATION

[Under section 32 of the Companies Ordinance, 1984 (XLVII of 1984)]

Corporate Universal Identification No.0094119

I hereby certify that **KK NETWORKS (PRIVATE) LIMITED** is this day incorporated under the Companies Ordinance, 1984 (XLVII of 1984) and that the company is **Limited by Shares**.

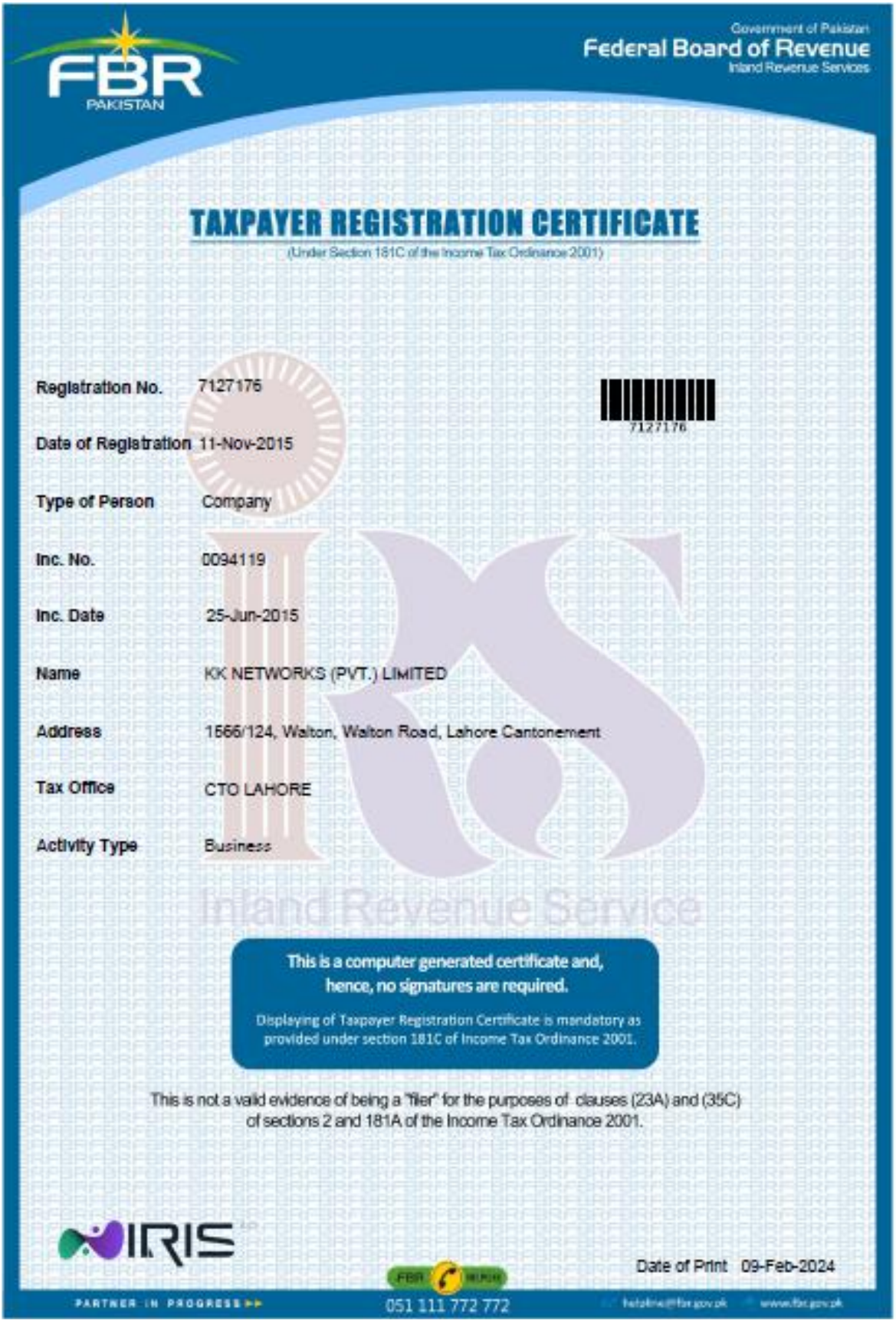
Given under my hand at **Lahore** this **Twenty Fifth** day of **June,** Two **Thousand** and **Fifteen**.

Fee Rs.**4,500/-**

(LIAQAT ALI DOLLA)
Additional Registrar

No.ARL/ 31659 DATED: 25/6/2015

CERTIFIED TRUE COPY SECP



Government of Pakistan
Federal Board of Revenue
Inland Revenue Services

FBR PAKISTAN

TAXPAYER REGISTRATION CERTIFICATE

(Under Section 181C of the Income Tax Ordinance 2001)

Registration No.	7127176
Date of Registration	11-Nov-2015
Type of Person	Company
Inc. No.	0094119
Inc. Date	25-Jun-2015
Name	KK NETWORKS (PVT.) LIMITED
Address	1666/124, Walton, Walton Road, Lahore Cantonement
Tax Office	CTO LAHORE
Activity Type	Business

7127176

Inland Revenue Service

This is a computer generated certificate and, hence, no signatures are required.

Displaying of Taxpayer Registration Certificate is mandatory as provided under section 181C of Income Tax Ordinance 2001.

This is not a valid evidence of being a "filer" for the purposes of clauses (23A) and (35C) of sections 2 and 181A of the Income Tax Ordinance 2001.

IRIS

Date of Print 09-Feb-2024

PARTNER IN PROGRESS

051 111 772 772

PUNJAB REVENUE AUTHORITY
GATEWAY TO DEVELOPMENT

Date : 30-07-2024
Time : 12:29:11

PNTN	**7127176-4**	Category	COMPANY
Name	KK NETWORKS (PVT.) LIMITED		

Business Name

Sr.	Business Name
1.	KK NETWORKS (PVT) LIMITED

CNIC/Reg No.	0094119
Address	HOUSE, 1566/124 WALTON WALTON ROAD , LAHORE , LAHORE CANTT
Block/Sector/Road	--- City LAHORE CANTT
Service Category	9815.6000 - SERVICES PROVIDED BY PROFESSIONALS AND CONSULTANTS ETC. / INFORMATION TECHNOLOGY-ENABLED OR INFORMATION TECHNOLOGY BASED SERVICES ,9815.6000
Date of Registration with PRA	9/7/2017 12:00:00 AM
Operational Status at PRA	Active

Print Back

Government of Pakistan
PAKISTAN TELECOMMUNICATION AUTHORITY
F-5/1, Islamabad
Tel: 051-9204455, Fax: 051-2878129, Email: tanweerkhattak@pta.gov.pk

No. PTA/Lic-WrL/LL/1825/2023 25th April, 2024

Subject: **Grant of Fixed Local Loop *(FLL)* License for WTR to KK Networks (Private) Limited**

Reference: *KK Networks (Private.) Limited's subject application dated 9th June, 2023.*

In pursuance to above referred application submitted for provision of Local Loop Telecommunication Services in Western Telecom Region (WTR), the Authority is pleased to inform that your application, based on provision of Mandatory Services, for grant of Fixed Local Loop (FLL) license for WTR is approved. The License shall be signed by Chief Executive Officer/ Authorized Representative of the applicant.

2. You are advised to submit the Initial License Fee amounting US$ 10,000/- or equivalent amount in Pak Rupees in the form of Bank Draft/Pay order in favor of Pakistan Telecommunication Authority within one month from issuance of this letter. The rate of exchange for determining the equivalent amount in Pak Rupees shall be the TT selling rate of National Bank of Pakistan prevailing at the business day proceeding the date of payment.

3. Furthermore, under this License, KK Networks (Private) Limited shall operate and provide all its authorized Mandatory Services that also includes those Class Value Added Services being provisioned in said Licensed Region i.e. WTR vide its Data CVAS (Internet/Data) License No. DIR(L)/CVAS-718/PTA/2015 dated 3rd November, 2015, for its respective region i.e. WTR after issuance of Commencement Certificate for this License.

4. KK Networks (Private) Limited, shall obtain prior approval of the Authority at the time it begins to offer a new category of Licensed Services, other than Mandatory Services.

5. This issues with approval of the Authority, please

Tanweer Shehzad Khattak
Director (Licensing)

To:

Mr. Imran Ali,
Chief Executive Officer,
KK Networks (Private.) Limited,
KK House, 1566/124, Walton Road,
Lahore Cantt.

1. **Director (Commercial Affairs), PTA HQs, F-5/1, Islamabad.**
2. **Director (Finance), PTA HQs, F-5/1, Islamabad.**
3. **Director (L&R), PTA HQs, F-5/1, Islamabad.**

Government of Pakistan
PAKISTAN TELECOMMUNICATION AUTHORITY
F-5/1, Islamabad
Tel: 051-9204455, Fax: 051-2878129, Email: tanweerkhattak@pta.gov.pk

No. PTA/Lic-WrL/LL/1824/2023 25th April, 2024

Subject: **Grant of Fixed Local Loop *(FLL)* License for STR-V to KK Networks (Private) Limited**

Reference: *KK Networks (Private.) Limited's subject application dated 9th June, 2023.*

In pursuance to above referred application submitted for provision of Local Loop Telecommunication Services in Southern Telecom Region-V (STR-V), the Authority is pleased to inform that your application, based on provision of Mandatory Services, for grant of Fixed Local Loop (FLL) license for STR-V is approved. The License shall be signed by Chief Executive Officer/ Authorized Representative of the applicant.

2. You are advised to submit the Initial License Fee amounting US$ 10,000/- or equivalent amount in Pak Rupees in the form of Bank Draft/Pay order in favor of Pakistan Telecommunication Authority within one month from issuance of this letter. The rate of exchange for determining the equivalent amount in Pak Rupees shall be the TT selling rate of National Bank of Pakistan prevailing at the business day proceeding the date of payment.

3. Furthermore, under this License, KK Networks (Private) Limited shall operate and provide all its authorized Mandatory Services that also includes those Class Value Added Services being provisioned in said Licensed Region i.e. STR-V vide its Data CVAS (Internet/Data) License No. DIR(L)/CVAS-718/PTA/2015 dated 3rd November, 2015, for its respective region i.e. STR-V after issuance of Commencement Certificate for this License.

4. KK Networks (Private) Limited, shall obtain prior approval of the Authority at the time it begins to offer a new category of Licensed Services, other than Mandatory Services.

5. This issues with approval of the Authority, please

Tanweer Shehzad Khattak
Director (Licensing)

To:

Mr. Imran Ali,
Chief Executive Officer,
KK Networks (Private.) Limited,
KK House, 1566/124, Walton Road,
Lahore Cantt.

1. **Director (Commercial Affairs),** PTA HQs, F-5/1, Islamabad.
2. **Director (Finance),** PTA HQs, F-5/1, Islamabad.
3. **Director (L&R),** PTA HQs, F-5/1, Islamabad.

Government of Pakistan
PAKISTAN TELECOMMUNICATION AUTHORITY
F-5/1, Islamabad
Tel: 051-9204455, Fax: 051-2878129, Email: tanweerkhattak@pta.gov.pk

No. PTA/Lic-WrL/LL/1823/2023 25th April, 2024

Subject: **Grant of Fixed Local Loop *(FLL)* License for STR-I to KK Networks (Private) Limited**

Reference: *KK Networks (Private.) Limited's subject application dated 9th June, 2023.*

In pursuance to above referred application submitted for provision of Local Loop Telecommunication Services in Southern Telecom Region-I (STR-I), the Authority is pleased to inform that your application, based on provision of Mandatory Services, for grant of Fixed Local Loop (FLL) license for STR-I is approved. The License shall be signed by Chief Executive Officer/ Authorized Representative of the applicant.

2. You are advised to submit the Initial License Fee amounting US$ 10,000/- or equivalent amount in Pak Rupees in the form of Bank Draft/Pay order in favor of Pakistan Telecommunication Authority within one month from issuance of this letter. The rate of exchange for determining the equivalent amount in Pak Rupees shall be the TT selling rate of National Bank of Pakistan prevailing at the business day proceeding the date of payment.

3. Furthermore, under this License, KK Networks (Private) Limited shall operate and provide all its authorized Mandatory Services that also includes those Class Value Added Services being provisioned in said Licensed Region i.e. STR-I vide its Data CVAS (Internet/Data) License No. DIR(L)/CVAS-718/PTA/2015 dated 3rd November, 2015, for its respective region i.e. STR-I after issuance of Commencement Certificate for this License.

4. KK Networks (Private) Limited, shall obtain prior approval of the Authority at the time it begins to offer a new category of Licensed Services, other than Mandatory Services.

5. This issues with approval of the Authority, please

Tanweer Shehzad Khattak
Director (Licensing)

To:

Mr. Imran Ali,
Chief Executive Officer,
KK Networks (Private.) Limited,
KK House, 1566/124, Walton Road,
Lahore Cantt.

1. **Director (Commercial Affairs),** PTA HQs, F-5/1, Islamabad.
2. **Director (Finance),** PTA HQs, F-5/1, Islamabad.
3. **Director (L&R),** PTA HQs, F-5/1, Islamabad.

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No. PTA/Lic-WrL/LL/1822/2023 25th April, 2024

Subject: **Grant of Fixed Local Loop (FLL) License for NTR-II to KK Networks (Private) Limited**

Reference: *KK Networks (Private.) Limited's subject application dated 9th June, 2023.*

In pursuance to above referred application submitted for provision of Local Loop Telecommunication Services in Northern Telecom Region-II (NTR-II), the Authority is pleased to inform that your application, based on provision of Mandatory Services, for grant of Fixed Local Loop (FLL) license for NTR-II is approved. The License shall be signed by Chief Executive Officer/ Authorized Representative of the applicant.

2. You are advised to submit the Initial License Fee amounting US$ 10,000/- or equivalent amount in Pak Rupees in the form of Bank Draft/Pay order in favor of Pakistan Telecommunication Authority within one month from issuance of this letter. The rate of exchange for determining the equivalent amount in Pak Rupees shall be the TT selling rate of National Bank of Pakistan prevailing at the business day proceeding the date of payment.

3. Furthermore, under this License, KK Networks (Private) Limited shall operate and provide all its authorized Mandatory Services that also includes those Class Value Added Services being provisioned in said Licensed Region i.e. NTR-II vide its Data CVAS (Internet/Data) License No. DIR(L)/CVAS-718/PTA/2015 dated 3rd November, 2015, for its respective region i.e. NTR-II after issuance of Commencement Certificate for this License.

4. KK Networks (Private) Limited, shall obtain prior approval of the Authority at the time it begins to offer a new category of Licensed Services, other than Mandatory Services.

5. This issues with approval of the Authority, please

Tanweer Shehzad Khattak
Director (Licensing)

To:
Mr. Imran Ali,
Chief Executive Officer,
KK Networks (Private.) Limited,
KK House, 1566/124, Walton Road,
Lahore Cantt.

1. **Director (Commercial Affairs),** PTA HQs, F-5/1, Islamabad.
2. **Director (Finance),** PTA HQs, F-5/1, Islamabad.
3. **Director (L&R),** PTA HQs, F-5/1, Islamabad.

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No. PTA/Lic-WrL/LL/1496/2021

23rd February, 2023

Subject: **Grant of Fixed Local Loop *(FLL)* License for KTR to KK Networks (Pvt) Ltd.**

Reference: *KK Networks (Pvt.) Ltd.'s Application for KTR dated 24th February, 2022*

In pursuance to above referred application submitted for provision of Local Loop Telecommunication Services in Karachi Telecom Region (KTR), the Authority is pleased to inform that your application, based on provision of Mandatory Services, for grant of Fixed Local Loop (FLL) license for KTR is approved. The License shall be signed by Chief Executive Officer/ Authorized Representative of the applicant.

2. You are advised to submit the Initial License Fee amounting US$ 10,000/- or equivalent amount in Pak Rupees in the form of Bank Draft/Pay order in favor of Pakistan Telecommunication Authority within one month from issuance of this letter. The rate of exchange for determining the equivalent amount in Pak Rupees shall be the TT selling rate of National Bank of Pakistan prevailing at the business day proceeding the date of payment.

3. Furthermore, under this License, KK Networks (Pvt.) Ltd., shall operate and provide all its authorized Mandatory Services that also includes Class Value Added (Internet/ Data) Services for its respective region i.e. KTR after issuance of its Commencement Certificate for this License.

4. KK Networks (Pvt) Ltd shall obtain prior approval of the Authority at the time it begins to offer a new category of Licensed Services, other than Mandatory Services.

5. This issues with approval of the Authority, please.

Tanweer Shehzad Khattak
Director (Licensing)

To:

Mr. Imran Ali,
Chief Executive Officer,
KK Networks (Pvt.) Ltd.,
714-A3, Jeff Heights, Main Boulevard,
Gulberg III, Lahore.

Copy to:

1. **Director (Commercial Affairs),** PTA HQs, F-5/1, Islamabad.
2. **Director (Finance),** PTA HQs, F-5/1, Islamabad.
3. **Director (L&R),** PTA HQs, F-5/1, Islamabad.

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Tel: 051-9204455, Fax: 051-2878129, Email: tanweerkhattak@pta.gov.pk

No. PTA/Lic-WrL/LL/1495/2021 23rd February, 2023

Subject: **Grant of Fixed Local Loop (FLL) License for ITR to KK Networks (Pvt) Ltd.**

Reference: *KK Networks (Pvt.) Ltd.'s application dated 24th February, 2022.*

In pursuance to above referred application submitted for provision of Local Loop Telecommunication Services in Islamabad Telecom Region (ITR), the Authority is pleased to inform that your application, based on provision of Mandatory Services, for grant of Fixed Local Loop (FLL) license for ITR is approved. The License shall be signed by Chief Executive Officer/ Authorized Representative of the applicant.

2. You are advised to submit the Initial License Fee amounting US$ 10,000/- or equivalent amount in Pak Rupees in the form of Bank Draft/Pay order in favor of Pakistan Telecommunication Authority within one month from issuance of this letter. The rate of exchange for determining the equivalent amount in Pak Rupees shall be the TT selling rate of National Bank of Pakistan prevailing at the business day proceeding the date of payment.

3. Furthermore, under this License, KK Networks (Pvt) Ltd shall operate and provide all its authorized Mandatory Services that also includes those Class Value Added Services being provisioned in said Licensed Region i.e. ITR vide its Data CVAS (Internet) License No. DIR(L)/CVAS-718/PTA/2015 dated 3rd November, 2015, for its respective region i.e. ITR after issuance of Commencement Certificate for this License.

4. KK Networks (Pvt.) Ltd., shall obtain prior approval of the Authority at the time it begins to offer a new category of Licensed Services, other than Mandatory Services.

5. This issues with approval of the Authority, please

Tanweer Shehzad Khattak
Director (Licensing)

To:
Mr. Imran Ali,
Chief Executive Officer,
KK Networks (Pvt.) Ltd.,
714-A3, Jeff Heights, Main Boulevard,
Gulberg III, Lahore.

Copy to:
1. **Director (Commercial Affairs),** PTA HQs, F-5/1, Islamabad.
2. **Director (Finance),** PTA HQs, F-5/1, Islamabad.
3. **Director (L&R),** PTA HQs, F-5/1, Islamabad.

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No. PTA/Lic-WrL/LL/1821/2023

2nd May, 2024

Subject: **Grant of Fixed Local Loop *(FLL)* License for NTR-I to KK Networks (Private) Limited**

Reference: *KK Networks (Private.) Limited's subject application dated 9th June, 2023.*

In pursuance to above referred application submitted for provision of Local Loop Telecommunication Services in Northern Telecom Region-I (NTR-I), the Authority is pleased to inform that your application, based on provision of Mandatory Services, for grant of Fixed Local Loop (FLL) license for NTR-I is approved. The License shall be signed by Chief Executive Officer/ Authorized Representative of the applicant.

2. You are advised to submit the Initial License Fee amounting US$ 10,000/- or equivalent amount in Pak Rupees in the form of Bank Draft/Pay order in favor of Pakistan Telecommunication Authority within one month from issuance of this letter. The rate of exchange for determining the equivalent amount in Pak Rupees shall be the TT selling rate of National Bank of Pakistan prevailing at the business day proceeding the date of payment.

3. Furthermore, under this License, KK Networks (Private) Limited shall operate and provide all its authorized Mandatory Services that also includes those Class Value Added Services being provisioned in said Licensed Region i.e. NTR-I vide its Data CVAS (Internet/Data) License No. DIR(L)/CVAS-718/PTA/2015 dated 3rd November, 2015, for its respective region i.e. NTR-I after issuance of Commencement Certificate for this License.

4. KK Networks (Private) Limited, shall obtain prior approval of the Authority at the time it begins to offer a new category of Licensed Services, other than Mandatory Services.

5. This issues with approval of the Authority, please

Tanweer Shehzad Khattak
Director (Licensing)

To:

Mr. Imran Ali,
Chief Executive Officer,
KK Networks (Private.) Limited,
KK House, 1566/124, Walton Road,
Lahore Cantt.

1. **Director (Commercial Affairs),** PTA HQs, F-5/1, Islamabad.
2. **Director (Finance),** PTA HQs, F-5/1, Islamabad.
3. **Director (L&R),** PTA HQs, F-5/1, Islamabad.

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No. PTA/Lic-WrL/LL/1818/2023 23rd April, 2024

Subject: **Grant of Fixed Local Loop *(FLL)* License for HTR to KK Networks (Private) Limited**

Reference: *KK Networks (Private.) Limited's subject application dated 9th June, 2023.*

In pursuance to above referred application submitted for provision of Local Loop Telecommunication Services in Hazara Telecom Region (HTR), the Authority is pleased to inform that your application, based on provision of Mandatory Services, for grant of Fixed Local Loop (FLL) license for HTR is approved. The License shall be signed by Chief Executive Officer/ Authorized Representative of the applicant.

2. You are advised to submit the Initial License Fee amounting US$ 10,000/- or equivalent amount in Pak Rupees in the form of Bank Draft/Pay order in favor of Pakistan Telecommunication Authority within one month from issuance of this letter. The rate of exchange for determining the equivalent amount in Pak Rupees shall be the TT selling rate of National Bank of Pakistan prevailing at the business day proceeding the date of payment.

3. Furthermore, under this License, KK Networks (Private) Limited shall operate and provide all its authorized Mandatory Services that also includes those Class Value Added Services being provisioned in said Licensed Region i.e. HTR vide its Data CVAS (Internet/Data) License No. DIR(L)/CVAS-718/PTA/2015 dated 3rd November, 2015, for its respective region i.e. HTR after issuance of Commencement Certificate for this License.

4. KK Networks (Private) Limited, shall obtain prior approval of the Authority at the time it begins to offer a new category of Licensed Services, other than Mandatory Services.

5. This issues with approval of the Authority, please

Tanweer Shehzad Khattak
Director (Licensing)

To:

Mr. Imran Ali,
Chief Executive Officer,
KK Networks (Private.) Limited,
KK House, 1566/124, Walton Road,
Lahore Cantt.

1. **Director (Commercial Affairs),** PTA HQs, F-5/1, Islamabad.
2. **Director (Finance),** PTA HQs, F-5/1, Islamabad.
3. **Director (L&R),** PTA HQs, F-5/1, Islamabad.

To,
Assistant Director
Directorate of Electronic Media & Publications (DEMP)
BF Building, Zero Point, Islamabad
Pakistan

Date: May 15, 2026

Subject: Affidavit / Undertaking Regarding Non-Blacklisting

This is to solemnly declare and affirm that **KK Networks (Private) Limited** has neither been blacklisted, debarred, nor declared ineligible by any Federal Government Department, Provincial Government Department, Autonomous Body, Semi-Government Organization, Public Sector Entity, Regulatory Authority, or Procurement Agency in Pakistan.

It is further declared that no proceedings regarding blacklisting or debarment are pending against the company before any court, authority, or public procurement agency.

KK Networks undertakes that all information and documents submitted in connection with EPADS registration/prequalification are true and correct. In case any information is found false or misleading, the company shall be liable for any action under applicable laws, rules, and regulations.

This undertaking is issued for submission to EPADS / PPRA for official use and record.

For and on behalf of
KK Networks (Private) Limited

Authorized Signatory

Name: Imran Ali

Designation: CEO

Company Stamp

KK House 1566/124 Walton Road Lahore Cantt, Lahore 05444-Pakistan. | Ph: +92 42 35761439-41 | UAN: 111-KKN-111
www.kknetworks.com.pk | info@kknetworks.com.pk

D

Relevant Experience & Past Performance

A. IPTV / Streaming / Broadcast Projects

Project 1	Electronic and Social Media Management and Monitoring
Client	**Naval College, Islamabad** Government of Pakistan
Year	Development and Deployment in 2025 (5 months) Support & Maintenance since March 2026
Description	• **Objective:** To ingest and monitor 36+ local and foreign television and radio channels from IPTV for breakings news, trends, coverage, etc. • **Core Tasks:** ○ Ingest 36 local and international TV channels and store them for an extended periods of time – 2 months and more. ○ Powerful search tools including full-text search, date/time filtering, and combined search criteria for precision results. ○ Insights into system usage including, clip creation metrics, and content analysis patterns. ○ Creation, archiving, searching, repurposing, and easy sharing of clips. • **Key Data Sources:** IPTV & Web • **Deliverable:** Turnkey deployment of complete and scalable on-premise media monitoring and digital coverage solution based on BesTel's in-house developed electronic and social media monitoring solution – **Analysys** (http://analysys.tech).

Project 2	Electronic and Social Media Management and Monitoring
Client	**Directorate of Electronic Media and Publications** Ministry of Information and Broadcasting Government of Pakistan
Year	Development and Deployment in 2014 Support & Maintenance from 2014 till 2023
Description	• **Objective:** To ingest and monitor 100+ local and foreign television and radio channels from satellite, IPTV and web for breakings news, trends, coverage, etc. • **Core Tasks:** ○ Ingest 100+ local and international TV and radio channels and store them for an extended periods of time – 6 months and more. ○ Powerful search tools including full-text search, date/time filtering, and combined search criteria for precision results.

	○ Insights into system usage including channel viewership, clip creation metrics, and content analysis patterns. ○ Easily create campaigns to track TV commercials. Get AI generated 99.9% accurate reports on time and frequency of ads. • **Key Data Sources:** IPTV, Satellite, Web, Radio, and prerecorded audio and videos recordings. • **Deliverable:** A comprehensive digital coverage report with a timeline of key events, screen time of prominent personalities, a map of connections, and assessments based on findings. The system also tracks advertisements and generates reports on each occurrence.

Project 3	**Electronic and Social Media Management and Monitoring**
Client	**Pakistan Telecommunication Corporation Limited (PTCL)** Ministry of Communications Government of Pakistan
Year	2014 – Development and Deployment 2014 – 2023 – Support & Maintenance
Description	• **Objective:** To ingest and monitor 100+ local and foreign television and radio channels from satellite, IPTV and web for breakings news, trends, coverage, etc. • **Core Tasks:** ○ Ingest 100+ local and international TV and radio channels and store them for an extended periods of time – 6 months and more. ○ Powerful search tools including full-text search, date/time filtering, and combined search criteria for precision results. ○ Insights into system usage including channel viewership, clip creation metrics, and content analysis patterns. ○ Easily create campaigns to track TV commercials. Get AI generated 99.9% accurate reports on time and frequency of ads. • **Key Data Sources:** IPTV, Satellite, Web, Radio, and prerecorded audio and videos recordings. • **Deliverable:** A comprehensive digital coverage report with a timeline of key events, screen time of prominent personalities, a map of connections, and assessments based on findings. The system also tracks advertisements and generates reports on each occurrence.

Project 4	FM & AM Radio Stream Monitoring, Analysis and Management
Client	**Private Sector** End user: Government of Pakistan
Year	Development and Deployment in 2025 (5 months) Support & Maintenance since Feb 2026
Description	• **Objective:** To ingest, record and monitor multiple FM/AM radio channels on premises for content monitoring, ad tracking, audio fingerprinting, broadcast compliance and historical archival. • **Core Tasks:** ○ Ingest 30+ FM/AM radio channels directly at client premises through dedicated radio receivers and IP capture nodes, with 24×7 unattended operation. ○ Continuous recording and archival of every ingested channel for extended retention periods, with rolling buffer and configurable per-channel retention windows. ○ Powerful search across the audio archive: full-text search across speech-to-text transcripts, date/time filtering, channel filtering and combined-criteria queries. ○ Speech-to-text transcription for Urdu and English broadcasts, indexed and searchable. ○ Audio Fingerprinting Engine: real-time fingerprinting of every aired second; reference library of ads, jingles, signature tunes and political content; instant match-and-alert on re-airing. ○ Ad Tracking & Campaign Reports: create campaigns to track radio commercials; AI-generated reports with 99.9% accuracy on time, frequency, channel and day-part of every ad airing. ○ Insights and analytics: channel-wise airtime, ad-spot occupancy, repeated content detection, content category breakdown and viewership / listenership patterns. ○ Clip generation: on-demand audio clip extraction for any time window for sharing with the client, and downstream stakeholders. ○ Audit trail and centralized logging for all administrative and operational actions. • **Key Data Sources:** Off the air, Satellite, Web, Radio, and prerecorded audio recordings. • **Deliverable:** A comprehensive digital coverage report with a timeline of key events, screen time of prominent personalities, a map of connections, and assessments based on findings. The system also tracks advertisements and generates reports on each occurrence.

Project 4	Electronic and Social Media Management and Monitoring
Client	**Ministry of Education** Government of Nigeria
Year	Development and Deployment in 2017 Support & Maintenance from 2017 till 2020
Description	• **Objective:** To ingest and monitor 100+ local and foreign television and radio channels from IPTV for breakings news, trends, coverage, etc. • **Core Tasks:** ○ Ingest 100+ local and international TV channels and store them for at least 12 months. ○ Powerful search tools including full-text search, date/time filtering, and combined search criteria for precision results. ○ Insights into system usage including, clip creation metrics, and content analysis patterns. ○ Creation, archiving, searching, repurposing, and easy sharing of clips. • **Key Data Sources:** IPTV, Radio & Web • **Deliverable:** Turnkey deployment of complete and scalable media monitoring and digital coverage solution based on BesTel's in-house developed electronic and social media monitoring solution – **Analysys** (http://analysys.tech). This was a cloud-based deployment.

Project 5	Integrated CCTV Surveillance & Smart TV Channel Monitoring with AI
Client	**Nine Tree Hotel** Lahore, Pakistan
Year	Development and Deployment in 2024 in 7 months Support & Maintenance since 2024 for the past 4 months
Description	• **Objective:** To deliver an integrated on-premise video intelligence solution for Nine Tree Hotel, Lahore covering **(a)** centralized CCTV surveillance of the entire hotel with browser-based video wall, recording, theft detection and motion detection, and **(b)** a dedicated in-room TV channel ingestion and monitoring facility for 30+ channels with video wall display and recording. • **Core Tasks – Part A: Hotel-Wide CCTV Surveillance:** ○ Integration with the hotel's existing IP CCTV camera infrastructure across all floors, lobby, reception, corridors, parking, kitchens and back-of-house areas.

	○ Centralized streaming of all CCTV camera feeds into BesTel's software for live operations and historical review. ○ Browser-native multi-layout video wall (1×1, 2×2, 3×3, 4×4 and larger configurable layouts) displaying live CCTV feeds in security operations center and reception. ○ Continuous 24×7 recording of every camera with configurable retention; rapid timeline scrubbing and clip extraction for incident review. ○ Theft Detection: AI-driven detection of suspicious behavior patterns — unattended objects, loitering, perimeter intrusion, and high-risk area access — with real-time alerts to the security operations center. ○ Motion Detection: zone-based motion detection per camera with configurable sensitivity, scheduling and after-hours intrusion alerts. ○ Search across recorded footage by camera, time window, motion-event or alert type. ○ Role-based access control: security staff, management and audit roles with separate permissions and full audit trail of every clip access. • **Core Tasks – Part B: Dedicated TV Channel Monitoring Room:** ○ Provisioning of a dedicated room within the hotel premises equipped with ingestion hardware to receive 30+ local and international television channels. ○ Real-time ingestion, transcoding, and centralized streaming of all 30+ channels into BesTel's software. ○ Browser-native multi-layout video wall display for the 30+ TV channels alongside the CCTV wall, allowing simultaneous viewing of any combination. ○ Continuous recording of every TV channel with configurable retention, rolling buffer and on-demand clip extraction. ○ Channel grouping, language tagging, and search across the recorded TV archive. ○ Single unified administration console for both CCTV and TV channel workflows. • **Key Data Sources:** IP CCTV Cameras (hotel-wide) and Television Channels (terrestrial / satellite / IP feeds in the dedicated monitoring room). • **Deliverable:** Turnkey deployment of a complete, scalable on-premise video surveillance, TV monitoring and AI-driven security intelligence solution at Nine Tree Hotel, Lahore, based on BesTel's in-house developed media intelligence platform — Analysys (**http://analysys.tech**).

B. API Content Delivery / Integration Projects

Project 1	**Threat Actor Profile (TAP) - "PhantomAccount"**
Client	A financial sector organization in the USA
Year	2019 – Development and Deployment 2020 – Support & Maintenance
Description	• **Objective:** To identify, profile, and monitor a specific threat actor ("PhantomAccount") known for selling stolen data on dark web forums. • **Core Tasks:** 1. Locate and archive "PhantomAccount's" posts and threads on key dark web markets. 2. Identify affiliated aliases or accounts on other platforms (X/Twitter, Telegram, code repositories). 3. Analyze the types of data being sold (e.g., financial, corporate, PII) to assess their focus and capability. 4. Monitor for new posts or data dumps to track ongoing activity. • **Key Data Sources**: Dark web forums (Dread, exploit.in), Telegram channels, X/Twitter, GitHub/Pastebin. • **Deliverable**: A consolidated intelligence profile document detailing aliases, TTPs (Tactics, Techniques, Procedures), and indicators of compromise (IoCs), updated bi-weekly. • **Relevance with CMU:** Data security, REST APIs, data storage and archival, and the video, audio and data analysis using extensive AI.

Project 2	**Brand and Intellectual Property (IP) Monitoring**
Client	**The Gap, UK**
Year	2017 – Development and Deployment 2017 – 2020 – Support & Maintenance
Description	• **Objective:** To map the complete digital footprint of "John Doe" to uncover hidden assets, affiliations, or deceptive behaviors for a due diligence check. • **Core Tasks:**

	○ Discover all known and unknown social media profiles (Facebook, Instagram, LinkedIn, etc.). ○ Uncover domain registrations, business affiliations, and property records under their name or known aliases. ○ Cross-reference email addresses and usernames against breach databases to find compromised accounts or reused passwords. • **Key Data Sources:** Social media platforms, WHOIS databases, public government records (e.g., SEC filings, property deeds), Have I Been Pawned, company registries. • **Deliverable:** A comprehensive digital footprint report with a timeline of key events, a map of connections, and a risk assessment based on findings. • **Relevance with CMU:** Intensive use of REST APIs, security over the web, extensive roles and role-based access.

Project 3	**Development of Auto Configuration Server (ACS) Software**
Client	**ALEA Solutions** (https://www.alea-soluciones.com/en/) Madrid, Spain
Year	2025 – Development and Deployment 2025 onwards – Support & Maintenance
Description	• **Objective:** Developed a modern Auto Configuration Server (ACS) compliant with TR-069 and TR-369 for centralized device provisioning, remote management, monitoring, and firmware control. • **Scope:** Implemented ACS functionalities including device onboarding, automated provisioning, remote diagnostics, firmware upgrades, and real-time device management with validation of 98 test cases across 5 DLD requirements. • **Architecture & Technologies:** Built using Node.js, PostgreSQL, Redis, and React with scalable backend services, real-time processing, and a web-based management dashboard. • **Key Features** ○ Remote device configuration and monitoring ○ Firmware and software upgrade management ○ Automated provisioning and diagnostics ○ Multi-device and multi-tenant support ○ Secure communication and operational logging

	• **Relevance:** Demonstrates expertise in carrier-grade telecom and IPTV infrastructure by enabling centralized management of routers, gateways, and IPTV set-top boxes for stable streaming, efficient troubleshooting, and large-scale device operations.

Project 4	**Development of Business & Operations Support Systems**
Client	**Zero Touch Technologies** Riyadh, Saudi Arabia
Year	2025 – On going Development and Deployment 45% work is finished and in Support & Maintenance Cycle 55% work is still under development
Description	• **Objective:** To develop a comprehensive and fully integrated suite of software solutions for both business and operations of small to medium telecom and internet service providers. • **Core Modules:** ○ Telco CRM Platform Ingest 100+ local and international TV and radio channels and store them for an extended periods of time – 6 months and more. ○ Powerful search tools including full-text search, date/time filtering, and combined search criteria for precision results. ○ Insights into system usage including channel viewership, clip creation metrics, and content analysis patterns. ○ Easily create campaigns to track TV commercials. Get AI generated 99.9% accurate reports on time and frequency of ads.
Core Modules	**1. Telco CRM Platform** • **Scope:** 13-milestone full-stack ISP CRM (React/Vite + Node.js + MongoDB) deployed at crm.bestel.tech. • **Scale:** 200 files, 120+ API endpoints, 22 Mongoose models. • **Relevance:** demonstrates capability to design, deliver and operate complex REST APIs at production scale — directly applicable to CMU's API tier. **2. Media Server & Log Management Server**

	• **Scope:** Media Server (Node.js + FFmpeg HLS multi-quality + React) and Log Management Server, deployed on Proxmox VMs with Docker Compose. • **Relevance:** demonstrates capability in HLS multi-quality streaming pipelines and centralized log management — both core to the CMU. **3. Audio Fingerprinting / Broadcast Monitoring System** • **Scope:** Ad-detection system for broadcast monitoring — Wang's Algorithm → SAF → ALISP+HMM, reaching 97% F1 score. • **Stack:** FastAPI backend with GPU acceleration (PyTorch). • **Relevance:** demonstrates broadcast-grade signal processing capability that can be layered on the CMU APIs in future phases.

Project 5	**Customized OSINT Solution**
Client	The Ministry of Home Affairs Government of Portugal
Year	2020 – Development and Deployment 2020 – 2022 – Support & Maintenance
Description	A powerful platform to quickly convert data overload into actionable intelligence. Our social media investigation solution was built to accelerate the entire lifecycle of cyber and public safety operations. • **Real-Time AI Monitoring:** Continuously tracks social media and public data. Our AI-powered sentiment analysis automatically flags potential threats and public safety disruptions as they emerge. • **Rapid Lead Generation:** Identifies and tracks the activities of suspects and threat actors to quickly uncover new leads and connections. • **Comprehensive OSINT Searches:** Runs fast, unified searches across a vast data ecosystem—including social media, the dark web, identity databases, and breach compilations. • **Relevance with CMU:** Data security, video analysis, digital forensics.

E

Client References & Performance Feedback

Government of Pakistan
Ministry of Information & Broadcasting
Directorate of Electronic Media & Publications
Benevolent Funds Building, Zero Point, Islamabad.

No: Emv/2021-PD
Tel No: 9252301
Fax No: 9252402

Date: 28-9-2021

To whom it may concern

This is to certify that in 2014 DEMP acquired the services of BesTel to develop, deploy, support, and maintain a state-of-the-art media and video management and search software. The software has the capability to ingest video streams from multiple television channels and process them into an intelligible system that can search and extract images and clips of interest. DEMP uses this cloud based system to track television commercials (TVCs), news and talk shows for their content.

The system is hosted at PTCL data centers in Islamabad and Lahore and makes use of PTCL's IPTV service. The project was successfully delivered to DEMP in 2015 after eight months of development and testing cycle. Ever since, BesTel is providing support and maintenance services for the software with occasional upgrades.

We are very pleased with the services of BesTel.

Amjad Rasheed
Project Manager

ptcl | BUSINESS SOLUTIONS

Date: November 28, 2025

To Whom It May Concern,

Subject: Experience Letter for BesTel Communications (Pvt) Ltd

This letter serves to formally acknowledge and appreciate the outstanding services provided by **BesTel** as our trusted technology partner for the design, implementation, and maintenance of a comprehensive Broadcast Monitoring and Analysis Solution.

In 2014, Pakistan Telecommunications Corporation Limited (PTCL) engaged BesTel to provide a **turnkey solution** for monitoring over 50 local and international television news channels. The project required a robust system capable of 24x7 recording, advanced media analysis, and detailed reporting.

BesTel successfully delivered an end-to-end solution, managing the entire project lifecycle with exceptional professionalism. Their services included:

a. **Requirements Collection & System Design:** Meticulous gathering of our needs and designing a scalable, custom architecture.

b. **Hardware Supply & Software Development:** Providing high-quality infrastructure and developing proprietary software tailored to our specific challenges.

c. **Deployment & Training:** Seamless system integration and comprehensive training for our operational staff.

The delivered solution exceeded our expectations. Its core capabilities included:

a. Continuous 24x7 recording of all designated channels.

b. Advanced search functionality for news coverage and specific topics of interest.

c. Analysis features such as screen time calculation, facial recognition, and transcription generation.

d. Optical Character Recognition (OCR) to read and analyze news tickers.

e. Generation of custom daily, weekly, and annual reports.

Following the successful deployment, BesTel provided complete support and maintenance services from **2014 to 2025**. Throughout this eleven-year period, their team demonstrated an unwavering commitment to reliability, responsiveness, and technical excellence, ensuring our system operated with maximum uptime and efficiency.

The BesTel solution has been an invaluable asset to PTCL, providing critical business intelligence and operational insights. We are thoroughly satisfied with the quality of their work, their problem-solving capabilities, and their dedication as a long-term partner.

We highly recommend BesTel for similar projects and wish them the very best in their future endeavors.

Sincerely,

Walayatullah
Senior Manager Pre-Sales (North)
Pakistan Telecommunications Corporation Limited (PTCL)

Pakistan Telecommunication Company Limited
PTCL Headquarters, Sector G-8/4, Islamabad-44000, Pakistan.
www.ptcl.com.pk/business PTCLBusiness PTCLBusinessSolutions

ADG LDI (PVT) LTD.

June 2, 2017

To whom it may concern

This is to certify we have procured **BizWyz ERP** from **BesTel.** We are currently using the following modules of BizWyz.

1. Procurement Management.
2. Warehouse Management.
3. Accounts and Finance Management.
4. Inventory Management.
5. HR Management.
6. Leave Management.
7. Time & Attendance.
8. Payroll Management.

We also declare that we are very happy and satisfied with **BizWyz** and the quality of support we get from them.

Additionally, please note that the total turnover of the company in year 2016 exceeded Rs. 200 Millions.

This certificate is being issued on the request of **BesTel Communications (Pvt) Ltd.**

Please feel free to contact the undersigned in case you have any queries.

Best regards.

LDI (PVT) LTD PAKISTAN

Kiran Bhatti
Manager HR
Cell#0301-8609986
Email.kiran.bhatti@adgldi.com

19-Babar Block, New Garden Town, Lahore - Pakistan. Tel: +92-42-35861543, 35940734 Fax : +92-42-35834021, 35940731

SPELL MOVIES™

Spell Movies LLC
106, Kirkland, WA 98083-0106
United States of America
Tel: +1-424-214-3170
info@spellmovies.com
www.spellmovies.com

June 5, 2017

To whom it may concern

This is to certify that Spell Movies LCC we have procured BizWyz ERP from BesTel. We are currently using the following modules of BizWyz.

1. Procurement Management
2. Warehouse Management
3. Accounts and Finance Management
4. Inventory Management
5. Online Store
6. HR Management
7. Leave Management
8. Time & Attendance
9. Payroll Management

We also declare that we are very happy and satisfied with BizWyz and the quality of support we get from them.

Additionally, please note that the total turnover of Spell Movies in year 2016 exceeded US$ 3.2 Million from its global operations.

This certificate is being issued at the request of BesTel Communications (Pvt) Ltd.

Please feel free to contact the undersigned in case you have any queries.

Best regards

Dr. Saqib Hassan Rathore
CEO

TRADITIONS

June 2, 2017

To whom it may concern

This is to certify that Traditions have procured BizWyz ERP from BesTel at our head office and at our four branches. We are using the following modules of BizWyz.

1. Procurement Management
2. Warehouse Management
3. Accounts and Finance Management
4. Inventory Management
5. Point of Sale (POS)
6. HR Management
7. Leave Management
8. Time & Attendance
9. Payroll Management

We also declare that we are very happy and satisfied with BizWyz and the quality of support we get from them.

Total Turnover of Tradition in year 2016 Rs=**242,334,202**.

This certificate is being issued at the request of BesTel Communications (Pvt) Ltd.

Please feel free to contact the undersigned in case you have any queries.

Best regards

Sohail Bajwa
General Manager
0300 865 3389
sohail.bajwa@traditions.com.pk

112/15 Quaid-e-Azam Industrial Estate Kot Lakhpat, Lahore, Pakistan.
Tel: 042-35114255-58,UAN: 042-111-TRD-111

www.traditionstores.com

QUAID-E-AWAM UNIVERSITY COLLEGE OF ENGINEERING,
SCIENCE & TECHNOLOGY, LARKANA – SINDH, PAKISTAN
Ph: 074-4055680, 4055670, 4167881

NO.20/QUCEST/LIB/Dated 20December, 2011

TO WHOM IT MAY CONCERN

This is to confirm that M/s Bes Tec Solutions (P.V.T) Ltd. Delivered us the required integrated library management software confirms to all requirements, specifications and formats.

This letter serves as the completion certificate, issued to M/S Bes Tec solutions (P.V.T) Ltd, at their request.

Assistant Librarian
Imtiaz Hussain Rahu

MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO

ISO-9000
CERTIFIED

No.PD(M-I&II)/MUET/JAM/2010/ 270 Dated 24 -09-2010

TO WHOM IT MAY CONCERN

This is certified that M/s Bestec has completed the work as per agreement, however some change requests formulated by the Librarian MUET, Jamshoro are under process, and will be completed in the end of December, 2010.

24/9/10

(SHABIR MAMONAI)
Project Director M-I&II

Address: Project Director M-I&II, Mehran University of Engineering & Technology, Jamshoro - 76062.
Phone #s: 022-2771212 (Off), 022-2772250-73/3011 (PABX), 022-2771403 (Fax), 0300-3444811 (Mob).
E-mail: mamonaishabir@yahoo.com.

QUAID-E-AWAM UNIVERSITY OF ENGINEERING, SCIENCE & TECHNOLOGY
NAWABSHAH-67480 SINDH PAKISTAN
Ph - (0244) 9370381-5 , Fax - (0244) 9370367
E-mail: information@quest.edu.pk Website: www.quest.edu.pk

No. QUEST/NH/LIB-234
Dated: 30th August 2010

TO WHOM IT MAY CONCERN

This is to confirm that on August 29, 2010, M/S BesTec Solutions (Pvt.) Ltd delivered us the required Integrated Library Management Software ***(ILM)***, in full and final format. The software conforms to all requirements, specifications, and formats.

This letter serves as the completion certificate, issued to M/S BesTec Solutions (Pvt.) Ltd, at their request.

Ghulam Farooque Channar

Incharge Librarian

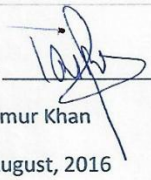
Customer Acceptance Form

Version 1.2 – Issue Date: August 16, 2016

Final Acceptance	
Client:	DMC, Lahore
Project:	BA Management Software for L'Oréal
Description:	Android based app for BA attendance, monitoring and sales reporting and stock management.

Acceptance of: ☐ Project Deliverable ☐ Project Phase ☑ Project

The project product (project deliverable, project phase or project) outcome has been measured against its prescribed acceptance criteria and has been formally accepted on behalf of the customer. The project may now be closed.

Executive / Sponsor:	Signature:  Name: Taimur Khan Date: 18 August, 2016
Project Manager:	Signature:  Name: Javariah Rasool Date: 18 August, 2016

F-1

Technical & Human Resource Capability

Project Team & Human Resources

BesTel will commit a dedicated team of full-time engineers and subject-matter experts to this project. The proposed team structure mirrors the human-resource expectations stated in Section 4(c) and Section 6(f) of the RFP.

A. Core Team Structure

Role	Headcount	Key Responsibilities	Qualification
Project Director (Sponsor)	1	Executive ownership, contract governance, escalation	Senior executive, 30+ years
Project Manager	1	Schedule, scope, risk, stakeholder reporting, change control	PMP/Prince2; CS/EE/IT; 7+ years
Solution Architect	1	End-to-end design, design authority, integration	CS/EE; 10+ years streaming/broadcast
Streaming / Broadcast Lead Engineer	1	Encoding, origin, failover, video wall	Senior engineer; 7+ years
System / Network Engineers	2	Network, VLANs, firewall, OS hardening	BSc + relevant certifications; 3+ years
Software / API Developers	2	REST APIs, SDKs, admin console, integrations	CS/SE; 3+ years
Front-End Engineer	1	Video wall, admin UI	CS/SE; 3+ years React
DevOps / SRE Engineer	1	CI/CD, Docker/K8s, monitoring, automation	BSc + cloud/DevOps certifications
Security Engineer	1	Hardening, WAF, IDS/IPS, pen-test	Security certifications (CEH/OSCP/CISSP)
QA / Test Lead	1	Test plan, FAT/SAT, automation	ISTQB or equivalent
Technical Writer	1	Documentation, training material	5+ years technical writing
NOC / Support Engineers (24×7 rota)	4+	Tier-1/Tier-2 support post go-live	Diploma/BSc; rotational shifts

B. Curriculum Vitae

Full CVs of the select proposed Project Director, Project Manager, Solution Architect, Streaming Lead, lead Software Developer and Security Engineer are attached on the following pages.

Waseem A. Rana

waseem.rana@bestel.tech
+92 300 841 2244

SENIOR EXECUTIVE / PROJECT DIRECTOR

Seasoned leader delivering fiscal, strategic, and operational solutions for uniquely challenging environments.

Strategic leader who translates analysis and insight into actionable improvements. Proficient in dissecting complex, cross-functional problems and leading teams to implement innovative solutions that benefit the bottom line. Builds alignment and drives change through exceptional interpersonal engagement, skilled negotiation, and a forward-thinking approach informed by a broad perspective.

Additional areas of expertise include:

- **Strategy & Governance:** Vision Planning | Policy Development | Regulatory Relations
- **Financial Leadership:** Budgeting & Cost Management | Profitability & P&L Analysis
- **Business Growth:** Sales & Marketing | Strategic Alliances & Negotiations
- **Operations & Talent:** Program/Product Management | HR | Team Building & Performance

EDUCATION

1991 - 1993	**M.Sc. Telecommunications, Controls & Digital Signal Processing** The University of Strathclyde, Glasgow, United Kingdom
1987 - 1991	**B.Sc. Electrical Engineering (Digital Systems)** Wichita State University, Wichita, Kansas, USA

SHORT COURSES

2014	**Management Essentials for Industry Professionals** The Institute for Career and Personal Development, LUMS, Lahore, Pakistan
2005	**Management Development Program** Lahore University of Management Sciences (LUMS), Lahore, Pakistan
2005	**Human Resource Management** University of The Punjab (PU), Lahore, Pakistan
1997	**Workshop on Advance Internet and Intranets** Lahore University of Management Sciences (LUMS), Lahore, Pakistan

WORK EXPERIENCE

Aug 05 to Present	**Founder & CEO** **BesTel Communications (Pvt.) Ltd., Pak &BesTel LLC, USA** **www.bestel.tech** BesTel is a global technology firm that develops and delivers advanced software and automation solutions for government and enterprise clients in telecommunications, finance, healthcare, and education.

The company operates on a unique venture-driven model, where individual products function as independent startups with dedicated leadership, teams, and full-cycle accountability for strategy, development, and go-to-market execution. This structure fosters agility and entrepreneurial focus within the larger organization.

In this dynamic environment, I hold full strategic and operational leadership, with direct P&L responsibility. My scope includes long-range planning, product lifecycle management, overseeing software development processes, and providing cross-functional executive oversight to drive business outcomes.

Apr 14 to Dec 17

CTO
Spell Movies (Pvt) Ltd, Pak & Spell Movies LLC, USA
www.spellmovies.com

Exclusive Home Entertainment Licensee for Major Studios.

- Held the exclusive licensing rights in Pakistan and Afghanistan for leading Hollywood studios, including Sony Pictures, Warner Bros., 20th Century Fox, Marvel Studios, and DC Comics.
- Engineered and launched a premium Video-on-Demand (VOD) service, partnering with major telecommunications providers to distribute licensed content to subscribers.
- Directed the end-to-end technology lifecycle: selected, developed, and integrated core systems for content delivery, DRM, backend infrastructure, and mobile applications while ensuring stringent Quality of Service (QoS) standards.

July 10 to July 11

Telecom / IT Specialist (Consultant)
National Disaster Management Authority (NDMA)
www.ndma.gov.pk

Appointed to architect critical disaster response systems, supported by World Bank funding. Led the technology evaluation and design for two major national initiatives: the **National Risk Assessment** and the **National Response System**.

My core mandate was to assess existing ICT infrastructure across federal, provincial (PDMA), and district (DDMA) levels and design integrated, scalable computer and telecommunications solutions. I developed comprehensive technical specifications to unify stakeholder operations—including Services HQ—enabling faster, more coordinated national disaster response.

Nov 02 to May 05

Senior Manager Projects
PCCW Global, Hong Kong
www.pccwglobal.com

One of Asia's leading integrated communications providers, delivering a full spectrum from local telephony to broadband and enterprise solutions.

- **Technical Leadership**: Directed the full lifecycle—planning, development, and implementation—of critical telecom infrastructure for voice/data communications, data streaming, and call management systems (IVR, voicemail).
- **Team & Operations Management**: Led a technical team through all facets of personnel management: assigning projects, evaluating performance, resolving issues, and overseeing schedules to ensure operational excellence.
- **Strategic Contribution**: Partnered with leadership to develop and implement program goals, policies, and strategic priorities, directly influencing departmental direction and governance.

Jan 00 to Oct 02

Project Manager, International Network Infrastructure
Passport Telecom, USA and UK
www.passporttele.com & www.passport-tele.com

Led business and technical execution for complex telecommunications deployments in the MENA region, supporting a global operator with a presence in 15+ countries.

- Served as a key technical lead for projects from conception to commission, designing and deploying SS7, ISDN, and VoIP networks.
- Managed the implementation of advanced access technologies including VSAT, SONET, and APON to build robust international communication infrastructure.
- Oversaw cross-functional project lifecycles, ensuring seamless integration of new platforms with the company's worldwide switching and network footprint.

Aug. 97 to Mar 99

Director Technical
Software (Pvt.) Ltd., Lahore, Pakistan
www.swl.com.pk

Led the technical strategy and business development for a pioneering IT services firm specializing in software development, network design, and mission-critical solutions.

- **Business & Technical Leadership:** Directed all project lifecycles, managing both technical execution and commercial strategy from initial tender preparation and client presentations through to final implementation and operational support.
- **Strategic Partnerships:** Leveraged and managed key international partnerships with industry leaders including Stratus Technologies (USA) for

fault-tolerant systems, Stratec Technologies (Singapore) for defense/security solutions, and Pernec Corporation (Malaysia) for systems integration.

- **Solution Delivery:** Owned end-to-end responsibility for deploying high-stakes IT solutions, ensuring technical excellence and alignment with client objectives across the public and private sectors.

July 94 to July 97 **Partner / Telecom Engineer**
Micro Electronics Technology Inc., Lahore, Pakistan

Directed engineering and production for Pakistan's largest professional radio manufacturer, specializing in land mobile radio (LMR) systems for high-security clients.

- **Strategic Technical Leadership:** Spearheaded the planning, design, and architecture of advanced wireless telecommunications systems and networks for mission-critical operations.
- **Operational Oversight:** Managed production and quality assurance departments, ensuring manufacturing excellence for HF, VHF, and UHF radio equipment.
- **Key Account Focus:** Engineered tailored communication solutions for the Pakistan Army—the company's primary client—as well as the Punjab/Sindh Rangers and other national law enforcement agencies.

May 95 to Jun. 00 **Visiting Faculty Member**

Delivered core curriculum and advanced coursework at multiple national and international institutions, including:

- **Foundation for the Advancement of Science & Technology (FAST)**
  Lahore, Pakistan
- **University of Central Punjab**
  Lahore, Pakistan
- **University of Nigeria**
  Nsukka, Nigeria

Oct. 93 to Jul. 94 **Research Assistant**
The Alan Turing Institute, Glasgow, UK
(www.turing.ac.uk)

I did research work on optimization of digital filters using genetic algorithms and neural networks.

NADEEM SHEIKH

Project Manager

Summary

IT Project Manager having 9+ years experience with background in Computer Science and MSc in Project Management, experienced in Agile-driven software delivery, stakeholder coordination, and end-to-end project execution with focus on timely and quality outcomes.

Work Experience

Project Manager, Bestel Communications (Pvt) Ltd, Lahore, Pakistan **March 2021 - Present**

- Managed software and IT projects from planning to deployment.
- Coordinated with development teams and clients to ensure timely project delivery.
- Monitored project progress, deadlines, and team performance.
- Gathered client requirements and ensured smooth communication between stakeholders and technical teams.
- Used Agile/Scrum methodologies to improve workflow and productivity.

Project Manager, Techlogix, Lahore, Pakistan **Aug 2018 - Feb 2021**

- Led end-to-end IT projects from planning to delivery using Agile/Scrum.
- Coordinated with developers, QA, and clients to ensure timely execution.
- Managed project scope, timelines, risks, and resource allocation.
- Gathered requirements and ensured clear communication between stakeholders. Delivered software solutions with a focus on quality and deadlines.

Junior Project Manager, Tkxel, Lahore, Pakistan **Jan 2017 - Apr 2018**

- Assisted in planning, execution, and delivery of IT projects using Agile/Scrum.
- Coordinated with developers, QA, and stakeholders to track progress and resolve issues.
- Supported project scheduling, documentation, and reporting.
- Ensured timely completion of tasks and smooth communication across teams.

Education

- **Master of Science in Project Management** — **2014-2016**
  Punjab University, Lahore, Pakistan — 3.36 CGPA

- **Bachelor of Science in Computer Engineering** — **2010-2014**
  UET, Lahore, Pakistan — 3.22 CGPA

Skills

- Project Planning
- Agile Methodologies
- Risk Management
- Budgeting & Forecasting
- Stakeholder Communication
- Resource Allocation
- Team Leadership

Hobbies

- Team Sports
- Reading Tech/Business articles
- Travelling
- Learning New technologies
- Strategy Games (Chess)
- Problem solving puzzles

NABEEL ANJUM

Software Engineer

Summary

Software Engineer with 5+ years of experience in designing, developing, and integrating scalable APIs for web and enterprise applications. Strong expertise in RESTful API design, system integration, and backend development, with hands-on experience building secure, high-performance services. Proven ability to work across full SDLC in Agile environments, collaborating with cross-functional teams to deliver reliable, cloudready solutions.

Technical Skills

- RESTful API Design & Development
- Microservices Architecture
- GraphQL API Design & Integration
- API Integration (Third-party services, Webhooks)
- API Security (OAuth2, JWT, API Keys)
- Backend Development (Node.js /Python / Java)
- Cloud Platforms (AWS / Azure /GCP)
- CI/CD Pipelines & Deployment

Professional Experience

API Developer Bestel Communications (Pvt) Ltd. Lahore, Pakistan Oct 2023 - Present

- Designed and developed RESTful APIs and integrated third-party services for web and enterprise applications.
- Worked on API security, microservices architecture, and system integration using Agile methodology.
- Collaborated with cross-functional teams to build scalable, cloud-ready backend solutions.

Software Engineer, Techverx May 2020 - April 2023

- Designed and developed RESTful APIs and backend services for web and enterprise applications.
- Integrated third-party APIs and optimized system performance for scalable solutions.
- Collaborated with cross-functional teams to deliver secure and reliable cloud-ready applications.

Education

- **BSc in Software Engineering** — **2015-2019**
  Comsats, Lahore, Pakistan — 3.19/4.0 CGPA
- **Intermediate - Pre Engineering** — **2013-2015**
  Punjab University — 87% Marks

Skills

- Photography
- Cycling
- Cricket

Mohammad Shahzeb Khan

Streaming Engineer

Summary

Streaming Engineer with 4+ years of experience in designing and supporting scalable streaming broadcast and network systems. Skilled in live video delivery, streaming protocols (HLS, DASH, RTMP) and optimizing end-to-end media pipelines for low latency and high reliability. Experienced in working with cloud infrastructure, CDNs and broadcast workflows to ensure stable, high quality streaming experiences.

Technical Skills

- Live Streaming Protocols (HLS, DASH, RTMP, WebRTC
- Video Encoding & Transcoding (FFmpeg, H.264/H.265)
- CDN & Content Delivery Optimization
- Low-Latency Streaming Systems
- Media Server Management
- Network & Streaming
- Troubleshooting

Professional Experience

Streaming Engineer – BesTel Communications (Pvt) Ltd. **May 2024 - Present**

- Designed and maintained scalable live and on-demand streaming systems for OTT/broadcast platforms.
- Worked with HLS, DASH, RTMP, and WebRTC for efficient low-latency video delivery.
- Optimized CDN performance, encoding pipelines, and media workflows to improve stream quality.
- Monitored system health and resolved streaming/network issues to ensure high availability and uptime.

Network Engineer, Cloudtech **2022 - 2024**

- Installed, configured and maintained LAN/WAN including routers, switches, and firewalls.
- Monitored network performance and resolved connectivity, latency, and security issues.
- Supported network troubleshooting, upgrades and ensured stable and secure system availability.

Internee, Streaming Engineer, Inara Technologies, Islamabad **2021 - 2022**

- Helped set up and test live streaming systems (HLS, RTMP).
- Assisted in video streaming, encoding, and quality checks.
- Supported fixing basic streaming and network issues with team.

Education

- **BSc., Computer Science** **2016-2020**
  FAST NUCES, Lahore, Pakistan 3.34/4.0 CGPA
- **Intermediate - Pre Engineering** **2014-2016**
  FC College, Lahore 81% Marks

Ahmed Raza

Cloud Engineer

Summary

Cloud Engineer with experience in designing, deploying, and managing scalable cloud infrastructure on AWS/Azure/GCP. Skilled in cloud automation, CI/CD pipelines, system monitoring, and security best practices. Strong ability to optimize performance, ensure high availability, and support reliable cloud-based applications.

Technical Skills

- AWS
- Azure
- GCP
- Terraform
- CI/CD Pipelines
- Kubernetes
- Docker
- API Integration
- Linux
- Git

Professional Experience

Cloud Engineer – BesTel Communications (Pvt) Ltd. **Jul 2024 - Present**

- Designed, deployed, and managed scalable cloud infrastructure on AWS/Azure/GCP.
- Automated CI/CD pipelines and cloud resource provisioning using modern DevOps tools.
- Monitored system performance, security, and availability to ensure high uptime and reliability.
- Collaborated with development teams to optimize cloud architecture and resolve production issues.

Network Engineer, KK Network, Lahore, Pakistan **Jan 2022 – Dec 2023**

- Managed cloud infrastructure resources including compute, storage, and networking for scalable applications.
- Implemented security best practices such as IAM policies, encryption, and access control management.
- Automated deployments and infrastructure provisioning using CI/CD pipelines and IaC tools.

Internee, Infrastructure, Contour Software, Lahore **Apr 2021 – Dec 2022**

- Assisted in deploying and managing cloud infrastructure on AWS/Azure, supporting application hosting and configuration.
- Monitored system performance and helped troubleshoot basic cloud, network, and deployment issues under senior guidance.

Education

- **BSc., Data Science** **Jan 2017 - Feb 2020**
  LUMS University, Lahore, Pakistan 3.49/4.0 CGPA

- **Intermediate - ICS** **2014-2016**
  GC University, Lahore 88.7% Marks

F-2

Technical Capability & Understanding of Requirements

Technical Capability & Understanding of Requirements

This section responds to the three sub-criteria stated by the Procuring Agency:

a. Understanding of IPTV acquisition modes, encoding, storage and performance requirements

b. Proposed high-level architecture for CMU streaming and API exposure; and

c. Consideration of security, scalability, redundancy, and 24×7 operations.

Each sub-criterion is addressed point-wise below with concrete, quantitative parameters drawn from BesTel's in-production Analysys platform, currently being used by multiple clients.

A. Understanding of IPTV Acquisition, Encoding, Storage & Performance

a. IPTV Acquisition Modes

- **Satellite (Ku-band / C-band):** dish + Integrated Receiver Decoder (IRD) for licensed satellite feeds; outputs ASI or IP (MPEG-TS over UDP).
- **IP / Multicast:** MPEG-TS over UDP/RTP multicast from telco aggregation networks, ingested via dual-NIC ingest servers with IGMPv3.
- **RTMP / RTSP Push:** for rights-holder push uplinks rather than pulls.
- **Hybrid acquisition by default** — the platform does not depend on any single bearer technology.

b. Encoding / Transcoding

- **Video codec:** H.264 (libx264) with the ultrafast preset and low-latency tuning; H.265/HEVC available where downstream consumers support it.
- **Audio codec:** AAC-LC, dual channel, 128 kbps default.
- **ABR ladder:** 1080p @ 4 Mbps / 720p @ 2.5 Mbps / 480p @ 1 Mbps — configurable per channel.
- **Segment length:** 4-second HLS segments (configurable down to 2 s for low-latency profiles).
- **Hardware acceleration:** optional NVIDIA NVENC or Intel Quick Sync for higher channel density per encoder node.
- **Output formats:** HLS, RTMP, MPEG-TS, and optionally Low-Latency HLS (LL-HLS) / LL-DASH.

c. Storage

- **Rolling buffer:** 30-day default per channel — configurable per channel and per category.
- **Container format:** fragmented MP4 (fMP4) aligned with HLS segments for seamless seek and clip extraction.
- **Capacity sizing:** approximately 120 TB raw for 124 channels at avg 3 Mbps with 30-day retention; **200–250 TB provisioned** for headroom and replication.
- **Storage backend:** clustered QNAP NAS (or equivalent) with RAID-6 and hot-spare drives; redundant 10GbE network paths.
- **Metadata store:** MongoDB replica set for channel/source metadata, events, RBAC, audit logs.

d. Performance Targets

Metric	Target
Platform availability	≥ 99.9%
Per-channel availability (excluding source-side outages)	≥ 99.5%
Failover switch-over time	≤ 2 seconds
Glass-to-glass latency (standard HLS)	≤ 6 seconds
Glass-to-glass latency (LL-HLS)	≤ 2 seconds
API p95 latency (metadata endpoints)	≤ 300 ms
API error rate (nominal load)	≤ 0.1%
Video wall cold-load	≤ 5 seconds
Recording continuity	≥ 99.9% segments persisted

B. Proposed High-Level Architecture — CMU Streaming & API Exposure

The CMU platform is structured as **seven horizontally-scalable tiers**, each independently deployable, replaceable and capable of being scaled out without architectural re-design.

#	Tier	Purpose
1	Acquisition Tier	Satellite IRD + IP/multicast endpoints with redundant network paths
2	Failover & Probing Tier	Continuous health probing of every source per channel; automatic switch-over to next healthy source
3	Encoding / Transcoding Tier	FFmpeg-based real-time transcoding to ABR rendition ladders
4	Streaming Origin Tier	MediaMTX cluster publishing HLS / RTMP / LL-HLS
5	API & Application Tier	Node.js + Express microservices delivering REST APIs, webhooks, video wall UI and admin console
6	Storage Tier	Rolling fMP4 recording archive (NAS) + MongoDB metadata store
7	Observability & Security Tier	Centralised ELK logging, Prometheus/Grafana metrics, WAF, IDS/IPS and audit trail

a. High-Level CMU Architecture (7 Tiers)

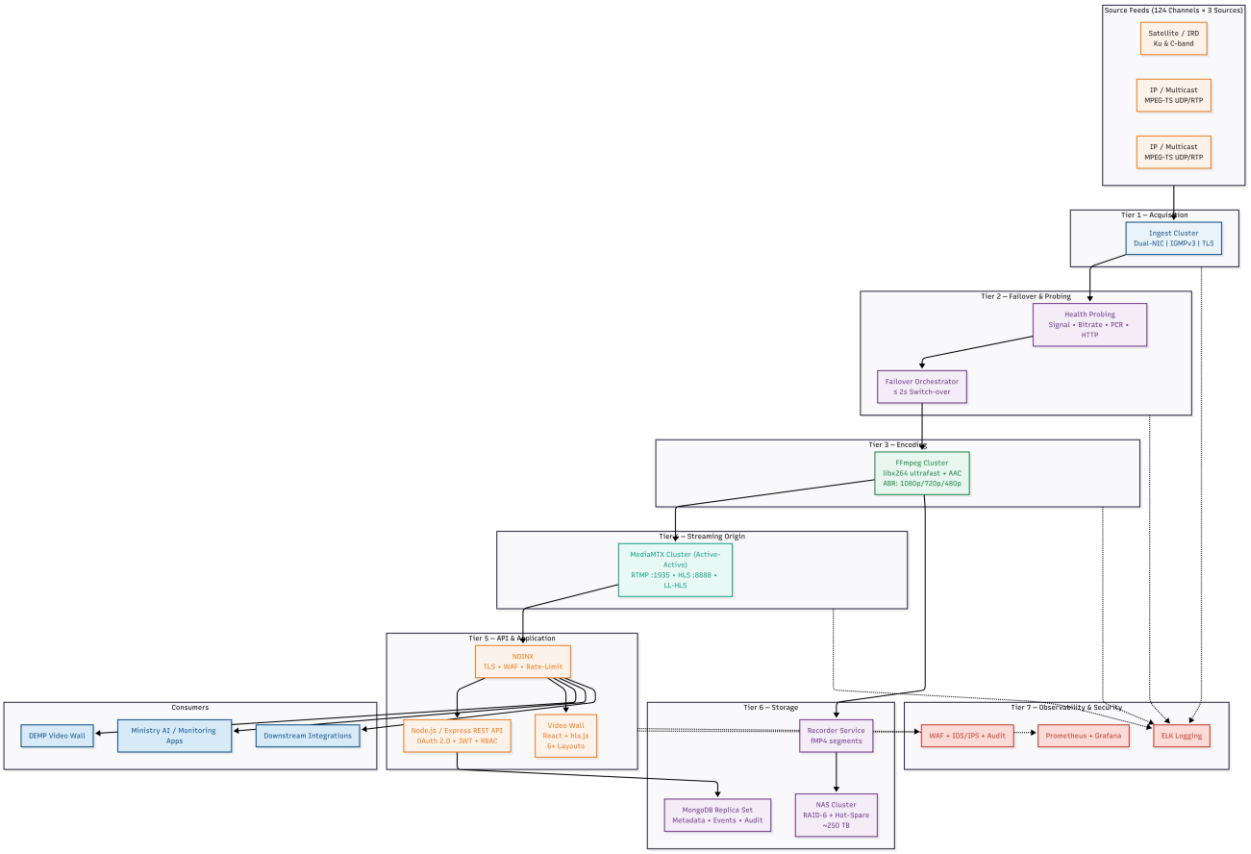


Figure 1 — CMU 7-Tier Architecture

b. End-to-End Data Flow (per channel)

- **Step 1 — Acquisition:** Source A (satellite/IRD), Source B (IP/multicast) and Source C (IP/multicast) deliver the live signal.
- **Step 2 — Health Probing:** each source is evaluated every 1–2 s against signal-presence, bitrate-floor, PCR continuity and (optionally) blank-frame heuristics.
- **Step 3 — Failover:** if the active source fails the probe, the orchestrator promotes the next healthy source within ≤ 2 s; downstream URL is unchanged.
- **Step 4 — Transcoding:** FFmpeg produces three ABR renditions per channel; segment length 4 s (default).
- **Step 5 — Publishing:** Transcoded streams are pushed via RTMP to MediaMTX origin (:1935) and republished as HLS (:8888).
- **Step 6 — Distribution:** authorized clients retrieve streams over HTTPS via the API gateway with auth, authorization and rate-limit.
- **Step 7 — Recording & Logging:** a recorder writes rolling segments to the NAS; every lifecycle event is logged centrally.

c. Per-Channel Data Flow with 3-Source Failover

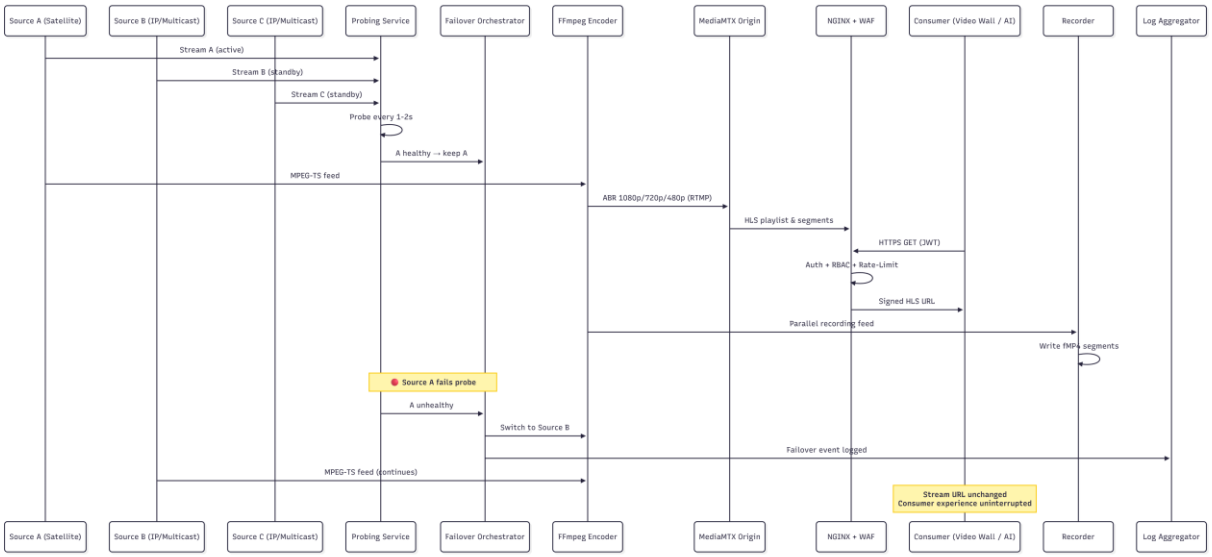


Figure 2 — Per-Channel Data Flow & Failover

d. API Exposure (REST + Webhooks)

- **GET /api/v1/channels** — list of all channels with metadata (id, name, language, category, region, current source, status).
- **GET /api/v1/channels/{id}/playlist.m3u8** — signed, authenticated HLS playlist URL.
- **GET /api/v1/channels/{id}/status** — real-time source, last failover, uptime and current bitrate.
- **GET /api/v1/recordings** — recorded segment lookup by channel and time window.
- **GET /api/v1/recordings/{id}/clip** — on-demand clip generation.
- **GET /api/v1/events** — auditable platform events (failovers, access, errors).
- **GET /api/v1/metadata/{channel}/transcript** — speech-to-text transcript stream.
- **Webhook subscriptions** — for failover events, content alerts and recording completions; HMAC-SHA256 signed payloads.
- **Developer artefacts** — OpenAPI 3.1 spec, Postman collection, Node.js and Python SDKs delivered at go-live.

e. Video Wall

Browser-native **React + Vite + hls.js** Single Page Application with **six or more configurable layouts** — 1×1, 2×2, 3×3, 4×4, 5×5, 6×4/8×4 large wall, and a PiP-focus mode. Layouts are saved per user, recalled on login. Closed captions, on-screen channel labels and a synchronized time-strip are configurable globally.

C. Security, Scalability, Redundancy, and 24×7 Operations

a. Security — Defence-in-Depth

- **Perimeter:** Next-Gen Firewall + IDS/IPS + WAF + geo-IP filtering + DDoS scrubbing.
- **Transport:** TLS 1.2+ on every HTTP/HTTPS path; HSTS; modern cipher suites only.
- **At-rest encryption:** AES-256 on recording storage and database backups.
- **Identity:** OAuth 2.0 + JWT; optional OIDC federation; MFA mandatory for administrative roles.
- **RBAC:** supports predefined roles (Viewer / Analyst / Operator / Admin / Auditor) & custom roles.
- **Rate Limiting:** per-tenant token-bucket limits per endpoint group; HTTP 429 with Retry-After.
- **URL Signing:** time-limited signed HLS playlist URLs prevent unauthorized re-sharing.
- **Hardening:** CIS-aligned OS hardening, audited kernel audit, fail2ban on management hosts.
- **Vulnerability Management:** monthly scans, quarterly patch cycle, emergency CVE patching.
- **Audit Trail:** immutable write-once logging for all administrative actions; exportable to DEMP audit authority.
- **Network Segmentation:** DMZ / Acquisition / Processing / Storage / Management / DR zones with east-west firewall rules.

b. Network Security Zones

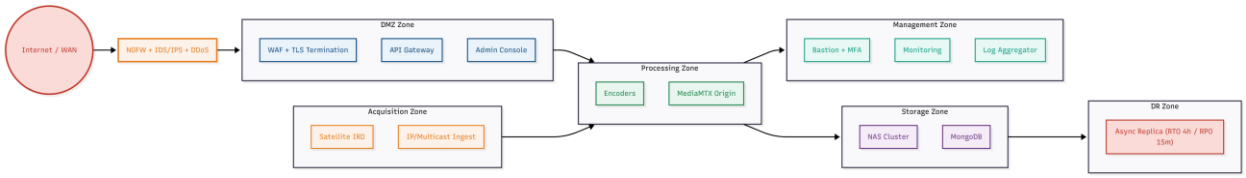


Figure 3 — Security Zones

c. Scalability

- **Horizontal scale-out** on every tier — ingest, encoder, origin or API nodes added without architectural redesign.
- **Stateless application tier** — services can be killed and re-instantiated without data loss; auto-scaling under load.
- **Caching tier (optional)** — CDN or internal cache layer for high fan-out distribution.
- **Capacity headroom** built into initial sizing for additional channels beyond 124.
- **Docker + Compose / Kubernetes** orchestration for consistent, repeatable scaling operations.

d. Redundancy

- **3-source failover per channel** with sub-2-second switch-over.
- **Active-active streaming origin** — each channel published to ≥ 2 MediaMTX nodes.
- **Active-active ingest** — dual ingest nodes with health-based traffic steering.
- **MongoDB replica set** (Primary + Secondary + Arbiter) for metadata HA.
- **NAS RAID-6 + hot-spare** for recording survivability.
- **Asynchronous DR replication** to a secondary site — RTO 4 hours; RPO 15 minutes (metadata) / 1 hour (recordings).
- **UPS + DG backup** at the CMU facility (on-premise mode).
- **Backup policy** — daily incremental, weekly full, quarterly tested restore drills.

e. 24×7 Operations

- **Pakistan-based NOC** manned 24×7 by Tier-1 / Tier-2 engineers.
- **Dedicated DEMP dashboard:** per-channel health, per-node metrics, API throughput, storage utilization and incident queue.
- **Preventive maintenance:** monthly OS patching, weekly health audits, quarterly DR drills.
- **Change management:** maintenance windows scheduled outside prime-time broadcast hours with advance notice to DEMP.
- **Monthly Service Review** with formal SLA report.

f. Incident Management SLA

Severity	Definition	Response	Resolution
S1 — Critical	Platform-wide outage / > 25 channels down	15 minutes	4 hours
S2 — High	1–25 channels down or API degraded	30 minutes	8 hours
S3 — Medium	Single source failure with failover active	2 hours	1 business day
S4 — Low	Cosmetic or feature request	1 business day	Next maintenance window

D. Compliance with Standards & Protocols

BesTel's proposed solution is built on **open IETF, ISO/IEC, ITU-T and W3C standards** exclusively. Every protocol used is publicly documented, interoperability-tested and supported by multiple independent implementations. There is no proprietary lock-in.

a. Broadcast & Streaming Standards

Standard	Reference	Use in CMU
MPEG-TS over UDP/RTP	ISO/IEC 13818-1	Primary satellite/IRD and telco multicast ingest
HLS	RFC 8216	Browser-native delivery and selected ingest
MPEG-DASH	ISO/IEC 23009-1	Optional ABR delivery
Low-Latency HLS / LL-DASH	Apple LL-HLS / DASH-IF	Sub-2-second latency profiles
RTMP / RTMPS	Adobe RTMP 1.0	Push uplinks and legacy delivery
RTSP	RFC 7826	IP camera/encoder uplinks
H.264 / AVC	ITU-T H.264, MPEG-4 Part 10	Primary video codec
H.265 / HEVC	ITU-T H.265	Optional high-density video codec
AAC-LC	ISO/IEC 14496-3	Audio codec
fMP4 / CMAF	ISO/IEC 14496-12, ISO/IEC 23000-19	Segment container, CMAF-compatible

b. API Standards

Standard	Reference	Use in CMU
REST architectural style	Fielding, 2000	Resource-oriented, idempotent verbs
OpenAPI Specification	OAS 3.1	Published, versioned, machine-readable API contract
JSON	RFC 8259	All request/response payloads
JSON Schema	draft 2020-12	Endpoint payload validation
HTTP/1.1 + HTTP/2	RFC 7230 / RFC 9113	Transport, HTTPS enforced
Webhook signing	HMAC-SHA256 (RFC 2104)	Signed webhook payloads to subscribers

c. Security Standards

Standard	Reference	Use in CMU
TLS 1.2 / 1.3	RFC 5246 / RFC 8446	Transport encryption on every external path
OAuth 2.0	RFC 6749	Token-based authentication
JWT	RFC 7519	Access token format
OpenID Connect (OIDC)	OIDC Core 1.0	Optional identity federation with DEMP IdP
Mutual TLS (mTLS)	RFC 8705	High-trust internal integrations
CIS Benchmarks	CIS Ubuntu / Docker / K8s	OS and container hardening baseline
OWASP Top 10 / API Top 10	OWASP 2023	Application security baseline
AES-256	FIPS 197	At-rest encryption
SHA-256	FIPS 180-4	Integrity & signing

d. Operational & Observability Standards

Standard	Reference	Use in CMU
Syslog	RFC 5424	Log forwarding to DEMP SIEM
Prometheus exposition format	Prometheus.io	Metrics scraping
OpenTelemetry	OTel 1.x	Distributed tracing
NTP / PTP	RFC 5905 / IEEE 1588	Clock synchronisation across nodes

E. Interoperability Considerations

a. Interoperability Map

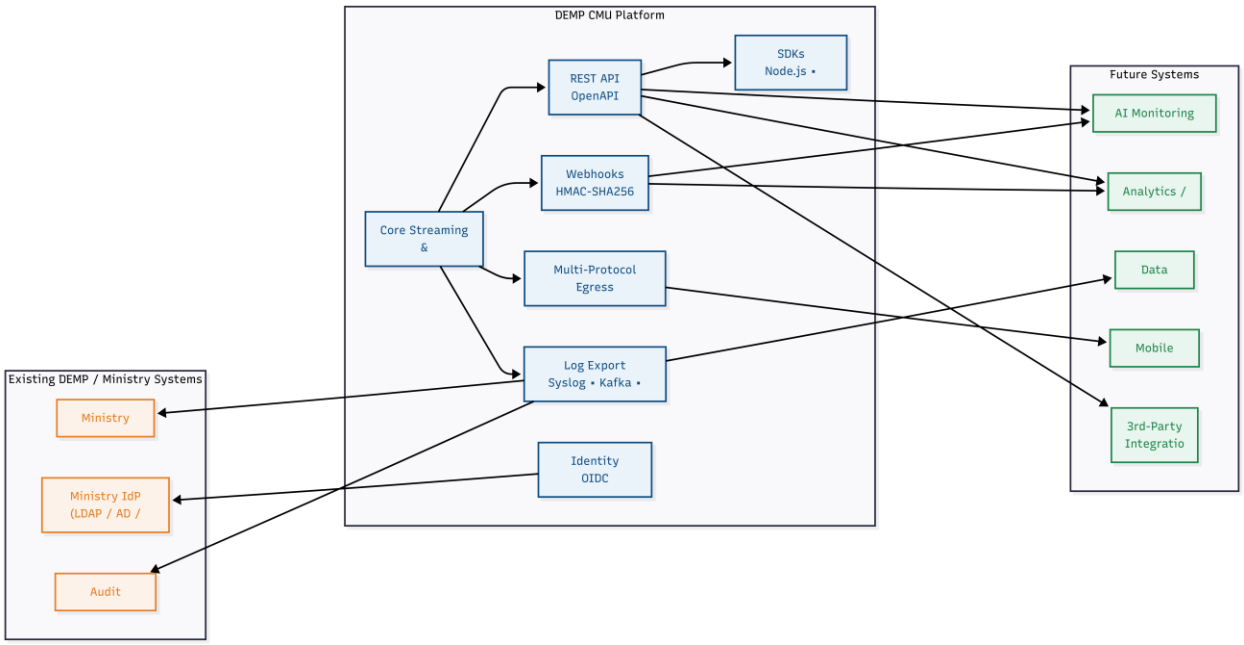


Figure 4 — Interoperability Map

b. Interoperability Considerations — Pointwise

- **Open-Standards Stack:** every protocol used is an open IETF / ISO / W3C / ITU-T standard — zero proprietary lock-in.
- **Vendor-Neutral Components:** FFmpeg, MediaMTX, NGINX, Node.js, MongoDB, Elasticsearch, Prometheus, Grafana — open source with active communities.
- **Hardware-Agnostic:** runs on any x86_64 server; not tied to any specific OEM.
- **Multi-Codec Output:** consumers receive H.264 (universal) or H.265 (where supported); ABR adaptation per device capability.
- **Multi-Protocol Egress:** the same channel can be delivered simultaneously as HLS, RTMP and LL-HLS for different downstream consumers.
- **API-First Design:** every platform function is exposed through the REST API — no functionality is locked inside the UI; downstream AI/monitoring apps can drive the platform programmatically.
- **SDKs Shipped:** Node.js + Python SDKs + Postman collection + sample integration code reduce downstream integration to hours.
- **Webhook Subscriptions:** push notifications eliminate polling overhead for downstream systems.

- **Syslog / Kafka / REST Log Export:** logs and events can be streamed to DEMP's existing SIEM, audit systems or data lake.
- **Identity Federation:** OIDC integration enables single sign-on with Ministry / DEMP existing identity providers.
- **Containerized Deployment:** Docker / Kubernetes packaging ensures identical operation on bare-metal, virtualized or cloud infrastructure.
- **Deployment Mode Flexibility:** the same platform can be delivered as on-premise, SaaS, or hybrid — DEMP retains full strategic choice without re-architecture.
- **Forward Compatibility:** the API is versioned (/api/v1/) with a deprecation policy; new versions co-exist with legacy versions for staged downstream migration.
- **Standards-Compliant Archive:** fMP4 / CMAF recording format is readable by any standards-compliant tooling — DEMP is never dependent on BesTel for archive access.

G

Financial Soundness & Stability

BesTel Communications (Pvt) Ltd.

May 14, 2026

To:
The Assistant Director
Directorate of Electronic Media & Publications (DEMP)
B.E Building, Zero Point
Islamabad, Pakistan

Subject: Average Annual Turnover Certificate

Dear Sir,

This letter is to certify that the current year-to-date turnover of BesTel Communications (Pvt) Ltd exceeds **Rs. 11,000,000 (Rupees Eleven Million).**

These figures are submitted as evidence of BesTel's financial capability and commitment to delivering high-quality results for the referenced project.

Thank you for your consideration.

Sincerely,





Waseem A. Rana
CEO
BesTel Communications (Pvt) Ltd
0300 841 2244
waseem.rana@bestel.tech

BesTel Communications (Pvt) Ltd.

May 14, 2026

To:
The Assistant Director
Directorate of Electronic Media & Publications (DEMP)
B.F Building, Zero Point
Islamabad, Pakistan

Subject: Net Worth and Financial Capacity Statement

Dear Sir,

This letter is to certify that BesTel Communications (Pvt) Ltd possesses sufficient financial strength and operational capacity to undertake and successfully execute DEMP's PSDP Project – Central Monitoring Unit for IPTV Streams.

The following figures are derived from the company's audited financial statements for the year 2025 and internal evaluations:

Description	Amount (PKR)
Shareholder's Equity / Net Worth	Rs. 3,080,000
Equipment and Hardware	Rs. 6,200,000
Software Products	Rs. 72,500,000
Total Assets-Based Net Worth	**Rs. 71,780,000**

These figures are submitted as evidence of BesTel's financial capability and commitment to delivering high-quality results for the project referenced.

Thank you for your consideration.

Sincerely,

mujtaba

BesTel Communications (Pvt) Ltd.

Mohammad Mujtaba
CRO
BesTel Communications (Pvt) Ltd
mohammad.mujtaba@bestel.tech

KKN

To,
Assistant Director
Directorate of Electronic Media & Publications (DEMP)
BF Building, Zero Point, Islamabad
Pakistan

Date: May 15, 2026

Subject: Average Annual Turnover Certificate

This is to certify that the annual turnover/revenue of KK Networks, as per audited financial statements, is as under:

- Revenue – FY 2025: Rs. 761,053,003
- Revenue – FY 2024: Rs. 1,614,473,803

Average Annual Turnover:
Rs. 1,187,763,403

The above figures are derived from the audited financial statements of the company and are submitted for EPADS registration/prequalification purposes.

For and on behalf of
KK Networks

Authorized Signatory

Name: Imran Ali

Designation: CEO

Company Stamp

KK House 1566/124 Walton Road Lahore Cantt, Lahore 05444-Pakistan. | Ph: +92 42 35761439-41 | UAN: 111-KKN-111
www.kknetworks.com.pk | info@kknetworks.com.pk

To,
Assistant Director
Directorate of Electronic Media & Publications (DEMP)
BF Building, Zero Point, Islamabad
Pakistan

Date: May 15, 2026

Subject: Financial Capacity

This is to certify that KK Networks (Private) Limited possesses sufficient financial resources and operational capability to manage and sustain its business operations and fulfill all obligations as required by your department.

For and on behalf of
KK Networks (Private) Limited

Authorized Signatory
Name: Imran Ali
Designation: CEO
Company Stamp

KK House 1566/124 Walton Road Lahore Cantt, Lahore 05444-Pakistan. | Ph: +92 42 35761439-41 | UAN: 111-KKN-111
www.kknetworks.com.pk | info@kknetworks.com.pk

To,
Assistant Director
Directorate of Electronic Media & Publications (DEMP)
BF Building, Zero Point, Islamabad
Pakistan

Date: May 15, 2026

Subject: Net Worth & Financial Capacity Statement

This is to certify that KK Networks possesses sufficient financial strength and operational capability based on its audited financial statements for the year 2025.

1. **Shareholders' Equity / Net Worth**:

   Rs. 235,258,899

2. **Asset-Based Net Worth (Operational Assets):**

   - Property, Plant & Equipment: Rs. 510,888,714
   - Intangible Assets / Licenses: Rs. 35,285,398
   - Stores & Spares: Rs. 226,327,695

   Total Asset-Based Net Worth: Rs. 772,501,807

The above figures are derived from the audited financial statements of the company and are submitted as evidence of the company's financial capacity for EPADS registration/prequalification purposes.

For and on behalf of
KK Networks (Private) Limited

Authorized Signatory

Name: Imran Ali

Designation: CEO

Company Stamp

KK House 1566/124 Walton Road Lahore Cantt, Lahore 05444-Pakistan. | Ph: +92 42 35761439-41 | UAN: 111-KKN-111
www.kknetworks.com.pk | info@kknetworks.com.pk

H

OEM / Technology Partnership & Approach

BesTel Communications (Pvt) Ltd.

May 14, 2026

To:
The Assistant Director
Directorate of Electronic Media & Publications (DEMP)
B.F Building, Zero Point
Islamabad, Pakistan

Subject: OEM / Principal Authorization

Dear Sir,

This letter is to certify that BesTel Communications (Pvt) Ltd is itself the Original Equipment Manufacturer (OEM) of **Analysys** (http://analysys.tech), a native video streaming / IPTV / encoder platform designed for ingestion, transcoding, recording, archiving, processing, analyzing, repurposing, searching, and sharing of TV and radio streams from multiple sources to ensure high availability and uptime.

As the OEM, BesTel Communications (Pvt) Ltd will be fully responsible and liable for providing the complete solution and all required services to DEMP as specified in the PSDP Central Monitoring Unit (CMU) for IPTV Streams Project.

Thank you for your consideration.

Sincerely,

Mohammad Mujtaba
CRO
BesTel Communications (Pvt) Ltd
mohammad.mujtaba@bestel.tech

Conclusion

The Central Monitoring Unit for IPTV Streams is a strategically important platform for the Ministry of Information & Broadcasting and the Government of Pakistan. Its long-term value depends not only on the initial build, but on the engineering depth, operational discipline and AI/analytics roadmap of the implementation partner.

BesTel Communications (Pvt) Ltd. respectfully submits that it is uniquely positioned to deliver this engagement:

- A production-grade IPTV intelligence platform (Analysys) is already operating with the Government of Pakistan at parallel scale.
- The proposed architecture is a battle-tested evolution of that platform — reducing risk and shortening time-to-go-live.
- The proposed timeline of 15 days to go-live + 7 days stabilization is realistic, measurable and underwritten by previous deliveries.
- BesTel maintains a Pakistan-based 24×7 NOC, full in-house engineering, and a clear AI roadmap (ContentSense, InstaClip, TVC Tracking, speech-to-text).
- The standards-first design (HLS, MPEG-TS over IP, REST/JSON, OAuth 2.0) guarantees that DEMP will never be locked into proprietary vendor technology.

BesTel thanks DEMP and the Ministry of Information & Broadcasting for the opportunity to participate in this pre-qualification, and welcomes the opportunity to demonstrate the platform, supply additional information, or appear before the evaluation committee at the Procuring Agency's convenience.

For and on behalf of BesTel Communications (Pvt) Ltd.



_____

Mohammad Mujtaba

Chief Revenue Officer

mohammad.mujtaba@bestel.tech

BesTel Communications (Pvt) Ltd.

https://bestel.tech